

Preset Horn–Anvil Gap as a Position-Triggered Displacement Constraint in Ultrasonic Metal Welding of Invar 36: Loading-Path Design for High-Strength, Low-Ductility Sheet Alloys

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Declaration of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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1 Preset Horn–Anvil Gap as a Position-Triggered Displacement Constraint in 2 Ultrasonic Metal Welding of Invar 36: Loading-Path Design for High-Strength, 3 Low-Ductility Sheet Alloys

4 Abstract

5 Ultrasonic metal welding of high-strength, low-ductility sheet alloys requires sufficient early-
6 stage interfacial activation while preventing late-stage thinning, overheating, and damage
7 accumulation. Using 0.53 mm Invar 36 sheets as a stringent model system, this study demonstrates
8 direct Invar–Invar ultrasonic metal welding and shows that a preset horn–anvil gap Δh can serve as
9 a position-triggered displacement constraint for loading-path design in pneumatic ultrasonic metal
10 welding. Once the lower dead point is reached, the process switches from force-dominant
11 indentation to displacement-constrained loading with static-force relaxation while vibration
12 continues. Coupled in-process monitoring, microstructural characterisation, and lap-shear testing
13 were used to resolve process–structure–property links. Decreasing the preset gap increased
14 collapse, force retention, late-stage tangential response, and thermal input, but not monotonically
15 in a beneficial manner. Three regimes emerged: large-gap under-loading, an intermediate robust
16 window, and zero-gap over-processing with persistent re-shearing. For the present condition set,
17 $\Delta h=0.5\text{--}0.7$ mm defined the robust window, and 0.7 mm was preferred because it eliminated
18 interface fracture at lower collapse and thermal demand than 0.5 mm and gave the most uniform
19 local mechanical response. By contrast, zero gap sustained late-stage re-shearing and promoted
20 seam damage despite higher collapse and higher power input. Joint performance was governed by
21 seam architecture and local compatibility rather than bonded area alone. A practical screening

requires the combination of termination power with collapse and force-retention descriptors. These results establish preset-gap scheduling as a displacement-centric framework for separating bonding-dominant and damage-dominant stages in ultrasonic welding of high-strength, low-ductility sheet alloys.

Keywords: Ultrasonic metal welding; Invar 36; Preset horn–anvil gap; Displacement constraint; Loading-path design; Weld collapse; High-strength, low-ductility sheet alloys

Introduction

Ultrasonic metal welding (UMW), as a solid-state joining process with short cycle times and comparatively low overall heat input, has long been regarded as an effective route for joining thin sheets and foils^[1,2]. For readily deformable metals such as aluminum, copper, and their alloys, UMW can often achieve rapid joining without bulk melting or a large heat-affected zone^[1,3]. However, once the target materials shift to high-strength, low-ductility metal sheets, obtaining stable high-quality joints becomes much more difficult. Under these conditions, parameter selection must satisfy two competing requirements simultaneously: sufficient interfacial shear and collapse are needed to disrupt surface films and establish continuous contact, but continued indentation and late-stage shear can rapidly introduce thinning, cracking, local overheating, and damage accumulation^[4,5]. The process window in such materials is therefore usually narrow, and weld quality is correspondingly more sensitive to fluctuation.

Existing literature still frames this problem mainly through machine-set parameters such as amplitude, normal load, weld time, and energy input^[1,2]. For the pneumatic UMW systems that dominate industrial practice, these quantities are essentially preset machine-side nominal settings that cannot usually be adjusted instantaneously in response to interface evolution during a single

weld. Kim et al. showed that, in thin dissimilar lap welding, joint quality is highly sensitive to deviations from the process window and that process robustness deteriorates markedly as the window narrows ^[6]. Liao et al. showed in a stainless-steel/Ti system that weld size is tightly coupled to local energy input, interfacial strength, and deformation level, so joint strength is not determined monotonically by nominal input alone ^[4]. Lin et al. likewise found in Ti/Ti welding that peak performance appears only within a limited intermediate regime, whereas further deformation instead triggers cracking and failure ^[5]. Taken together, these studies indicate that, for high-strength, low-ductility sheet metals, the key issue is not simply the nominal settings themselves, but how the thermomechanical loading path that governs the competition between bonding and damage evolves during welding and how that path can be effectively constrained.

For this reason, an increasing number of studies have turned toward internal process responses and in-situ monitoring quantities in order to characterize welding state more directly. Feng et al. systematically reviewed the signal types most commonly used in UMW quality monitoring, including force, power, displacement/collapse, vibration/acoustics, and temperature, and summarized the main strategies based on thresholding, feature extraction, and pattern recognition^[7]. At the application level, Lee et al. correlated online machine-side force, displacement, and power signals with weld attributes ^[8], whereas Müller et al. used vibration- and indentation-depth-related features to identify disturbed welds ^[9]. Balz et al. further combined structure-borne sound, laser vibrometry, and high-speed imaging to show that tool vibration and workpiece oscillation under different manufacturing conditions map systematically onto joint-quality variations ^[10]. Meanwhile, Lu et al. and Li and Balle, through in-situ relative-motion measurement and laser velocimetry, revealed the stage-wise evolution of bond formation during ultrasonic welding ^[11,12]. These studies show that the enclosed interface state can be tracked through

67 machine-side and in-process signals. Most of the existing work, however, still treats these signals
68 primarily as descriptors or screening tools rather than as design handles for constraining the late-
69 stage loading path itself.

70 Among these internal responses, collapse deserves to be singled out not merely because it is
71 easy to measure, but because it lies closest to the state transition between bond development and
72 over-processing. Bakavos and Prangnell demonstrated in high-power ultrasonic welding of
73 aluminum sheet the continuous evolution from micro-joint formation to interface expansion,
74 vortex-like flow, and shear-band development, showing that collapse is intimately related both to
75 joint formation and to excessive deformation ^[3]. Liao et al. on weld-size/energy-input coupling and
76 Lin et al. on the intermediate window in Ti/Ti joints likewise showed that moderate collapse
77 promotes the establishment of a continuous load-bearing interface, whereas once collapse
78 continues under sustained normal compression and late-stage shear, the process becomes much
79 more likely to enter thinning, cracking, local overheating, and damage accumulation ^[4,5]. Collapse
80 should therefore not be understood merely as a final height change or a simple quality descriptor; it
81 is better viewed as a core response marking process-state transition. Seen from this perspective,
82 existing displacement-related strategies do offer important inspiration. Abbas et al. introduced
83 additional root gaps in multi-wire ultrasonic joining to change the initial bonding geometry ^[13].
84 Baboi and Grewell used final height/post-height as a distance-based stop criterion, so that
85 ultrasonic vibration was terminated once a target height was reached ^[14]. More broadly, servo-
86 driven ultrasonic joining systems have shown the potential to manage loading history through
87 motion trajectory control in polymer/composite joining ^[15]. These approaches alter initial
88 geometry, stop timing, or active trajectory control, but they still do not answer the more specific
89 question that matters here: within a displacement-related control strategy, can a simple and

physically interpretable boundary-condition switch be triggered while sonics is still ongoing and the weld has not yet been terminated, so that the late-stage loading path is deliberately reshaped?

In the present work, Invar 36 is chosen as a stringent model system for this question. The low-expansion behavior of Fe–Ni Invar originates in the classical Invar effect ^[16]. Its functional response is clearly sensitive to thermal history and processing state ^[17,18]. Existing studies on Invar joining remain concentrated mainly on fusion-based routes. Zhan et al. compared hybrid laser–MIG and conventional MIG joining of Invar 36 ^[19]. Liu et al. showed in keyhole TIG welding that heat input can strongly alter the weld microstructure and properties of Invar 36 ^[20]. Further post-weld heat treatment can even change linear-expansion behavior ^[18]. For thin-sheet Invar systems with explicit dimensional-stability requirements, the issue is therefore not merely whether sound bonding can be achieved, but whether reliable joining can be established without introducing unnecessary late-stage indentation and thermomechanical overload. Invar is not used here as a simple material-specific case; rather, it acts as a stringent model system that magnifies a generic loading-path problem by superposing high strength, limited ductility, and a functional constraint that is highly sensitive to thermal history. To the authors’ knowledge, this study also provides the first direct peer-reviewed demonstration of Invar–Invar ultrasonic metal welding, while using that demanding system to extract a broader loading-path-design insight.

Against this background, the literature has already clarified which machine settings influence the process window and which in-process signals can indicate stage evolution. The more fundamental question is whether, in pneumatic metal welding, the late-stage loading path can be deliberately reshaped before welding is terminated, so that the deformation stage that promotes bonding is separated as far as possible from the over-processing stage that undermines joint integrity. The present study addresses this knowledge gap by examining whether a preset horn–

113 anvil gap can trigger a boundary-condition switch during welding and thereby optimize the loading
114 path. More specifically, the preset horn–anvil gap (Δh) is not treated here as a simple limit on final
115 displacement. Rather, under a prescribed static load, it defines the lower dead point that the horn is
116 allowed to reach; once that position is engaged, welding does not stop, but instead shifts from
117 force-dominant indentation to displacement-constrained loading with accompanying static-force
118 relaxation. Its research value therefore lies not in introducing yet another geometric parameter, but
119 in providing a simple, repeatable, and pneumatic-sheet-welding-compatible means of late-stage
120 path shaping: strong loading is retained initially to promote early bonding, whereas once collapse
121 and contact development have reached a sufficient level, a position-triggered boundary-condition
122 switch can suppress late-stage over-indentation and over-welding.

123 On this basis, the study addresses three interrelated questions: first, under fixed amplitude,
124 nominal normal load, and weld time, how does Δh change the timing and extent of static-force
125 relaxation; second, how does this late-stage path change influence weld-seam continuity,
126 microstructural refinement, local mechanical heterogeneity, and fracture-mode transition; third,
127 can these machine-side signatures be compressed into in-process indicators that are more
128 discriminating than termination power yet explicitly condition-dependent? Through a demanding
129 direct Invar–Invar welding demonstration, combined with in-situ monitoring, near-interface
130 temperature measurement, lap-shear testing, SEM/EBSD characterization, and selected-line
131 nanoindentation, the present work establishes the preset horn–anvil gap as a position-triggered
132 displacement constraint and extracts from it a process-design insight that is relevant beyond Invar
133 to pneumatic UMW of other high-strength, low-ductility sheet metals.

2. Materials and methods

2.1. Materials

An annealed Fe-36Ni Invar 36 sheet (Dongguan Chenhua Metal Materials Co., Ltd.) with a thickness of 0.53 mm was used as the base material. Rectangular coupons (8.0 mm × 50.0 mm) were cut with their long axis parallel to the rolling direction (RD). The chemical composition is listed in Table 1. Before welding, the coupons were ultrasonically cleaned in ethanol to remove loose surface contamination; no mechanical grinding was applied so that bonding developed from the as-received surface condition.

Table 1. Chemical composition of Invar 36 alloy.

Element	C	Mn	Si	S	P	Ni	Cu	Fe
Content (wt %)	0.005	0.43	0.12	0.001	0.001	36.03	0.01	Balance

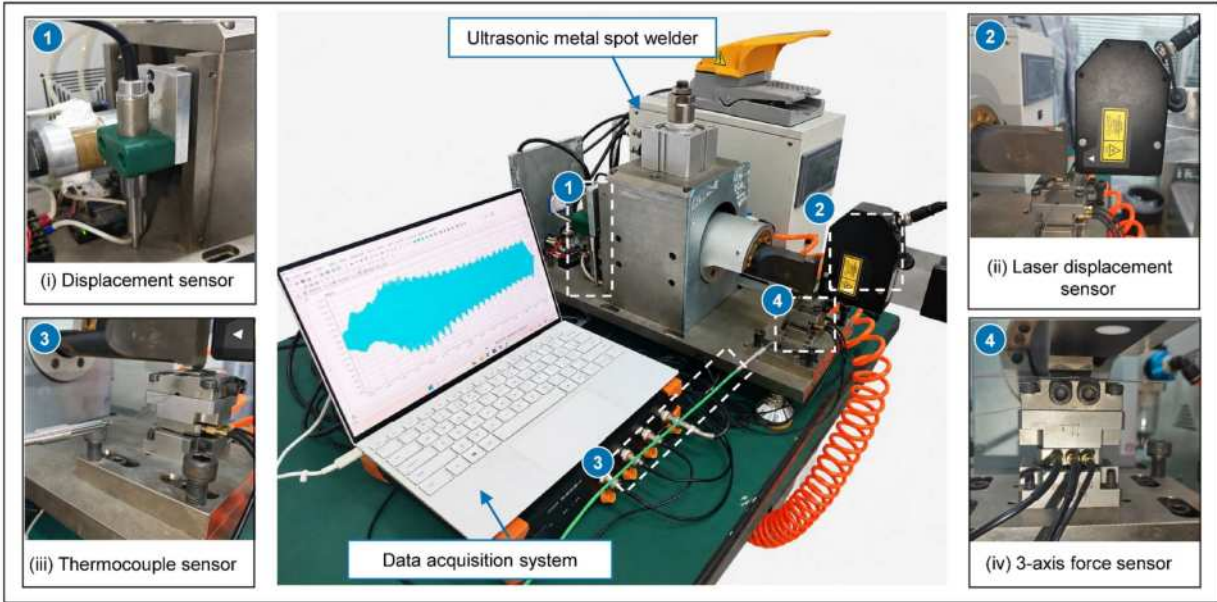
2.2. Experimental setup, process monitoring and welding procedure

2.2.1. Welding equipment and in-situ monitoring

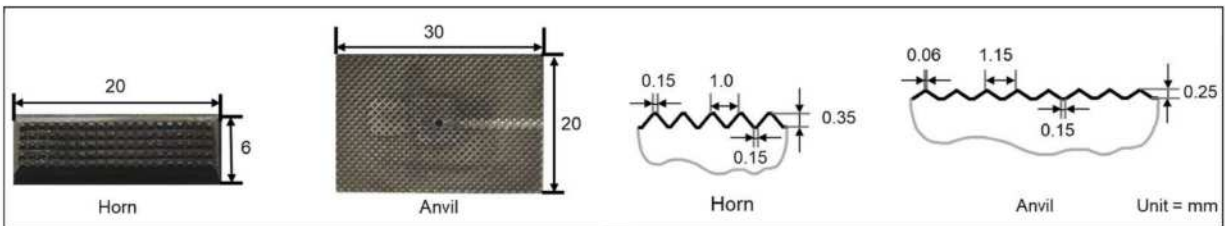
Ultrasonic spot welding was performed on a 3kW pneumatic ultrasonic metal welder equipped with a knurled horn and anvil (Fig. 1) using a 5 mm lap overlap. Two coordinate systems are used throughout this work. X-Y-Z denotes the machine-fixed sensing frame used for process monitoring, whereas RD-TD-ND denotes the specimen frame used for metallographic and crystallographic analysis. In the present setup, X is parallel to RD, Y is parallel to TD and is also the nominal vibration direction, and Z is parallel to ND. Tri-axial forces F_x , F_y , and F_z were measured by a three-axis high-frequency piezoelectric sensor installed beneath the anvil. These signals represent resultant machine-side reaction forces transmitted through the horn-sheet-anvil-fixture path and are therefore used as comparative process-monitoring signatures rather than as direct measurements of the local interfacial stress state. Horn displacement $u_{horn,z}$ was measured by

155 an inductive sensor mounted on the horn holder and was interpreted as the net reduction in stack
 156 thickness during welding. Horn vibration amplitude was monitored by a laser displacement sensor.
 157 A K-type thermocouple was inserted through an L-shaped hole in the anvil and positioned in a
 158 through-hole drilled at the weld centre of the lower sheet, so that the junction lay close to the faying
 159 interface without being placed between the sheets. The drilled specimens were used only for
 160 temperature measurement; all mechanical testing and microstructural characterization reported
 161 here were conducted on intact specimens welded under the same process parameters. The
 162 ultrasonic generator output power was recorded from the welder controller. Force and
 163 displacement signals were acquired in the same weld, whereas temperature and horn-amplitude
 164 data were collected from nominally repeated welds. Signals were sampled by an Arizon-DASP
 165 system at 1 MHz, while the horn-amplitude signal was sampled at 160 kHz. Temperature is
 166 reported as mean $\pm$ standard deviation from three repeated welds. Complete synchronized force-
 167 displacement records were available for a monitored subset ($n = 10$ per Δh) because occasional
 168 sensor-plug loosening occurred during high-frequency acquisition; repeatability statistics of
 169 monitoring-derived features were therefore obtained from this subset.

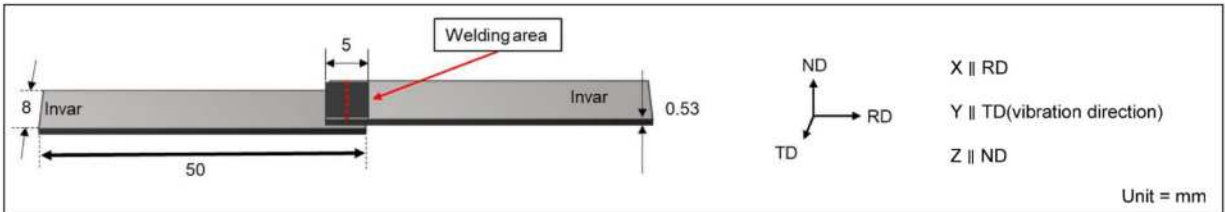
170 For data reduction, the measured force and displacement signals were decomposed into
 171 slowly varying and oscillatory components. The static normal force $F_{Z,static}$ was obtained by cycle
 172 averaging (equivalent to low-pass filtering) of F_Z . For the force signals, the dynamic force-
 173 envelope amplitudes $\tilde{F}_x$, $\tilde{F}_y$, and $\tilde{F}_z$ were quantified as the corresponding half peak-to-peak
 174 values. The upper and lower envelopes were extracted from the full raw records using a 0.5 ms
 175 moving window with a 0.25 ms step.



(a) Overall experimental setup with sensor locations



(b) Horn and anvil textures and profile dimensions.



(c) Lap-specimen dimensions and coordinate definition.

176

Fig. 1. Experimental setup and measurement system for ultrasonic metal welding of Invar 36. (a) Overall ultrasonic metal spot welding setup integrated with the DASP data acquisition system, with enlarged sensor insets and numbered callouts identifying the key monitoring components: (i) displacement sensor for real-time monitoring of horn displacement; (ii) laser displacement sensor for monitoring horn vibration amplitude during welding; (iii) K-type thermocouple inserted through the anvil and positioned at the weld centre near the faying interface via a pre-drilled hole in the lower sheet; and (iv) tri-axial high-frequency piezoelectric force sensor for monitoring three-dimensional forces during ultrasonic welding. (b) Horn and anvil textures and profile dimensions. (c) Lap-specimen dimensions and coordinate definition. RD, TD, and ND denote the rolling,

transverse, and normal directions, respectively. In the process-monitoring coordinate system, $X \parallel RD$, $Y \parallel TD$ and corresponds to the vibration direction, and $Z \parallel ND$.

2.2.2. Welding procedure, parameters and preset horn-anvil gap Δh

The primary process variable was the preset horn-anvil gap Δh , defined here as the minimum horn-anvil separation at which the preset static normal load F_0 is still maintained. Once the horn moves below this position during deformation, the mechanical stop constrains further downward stroke, the static load begins to relax, and ultrasonic vibration continues until the prescribed weld time is reached. In the present pneumatic system, Δh therefore acts as a mechanically preset, position-triggered displacement constraint rather than as an ultrasonic stop criterion.

Δh was set using a double-nut mechanical stop as summarized in Fig. 2. After the reference static load had been established, a feeler gauge equal to the target Δh was inserted, the stop position was adjusted until the monitored static load again matched the reference value, and the nuts were locked. The setting was then verified with slightly thicker and thinner gauges to confirm engagement close to the intended lower dead point.

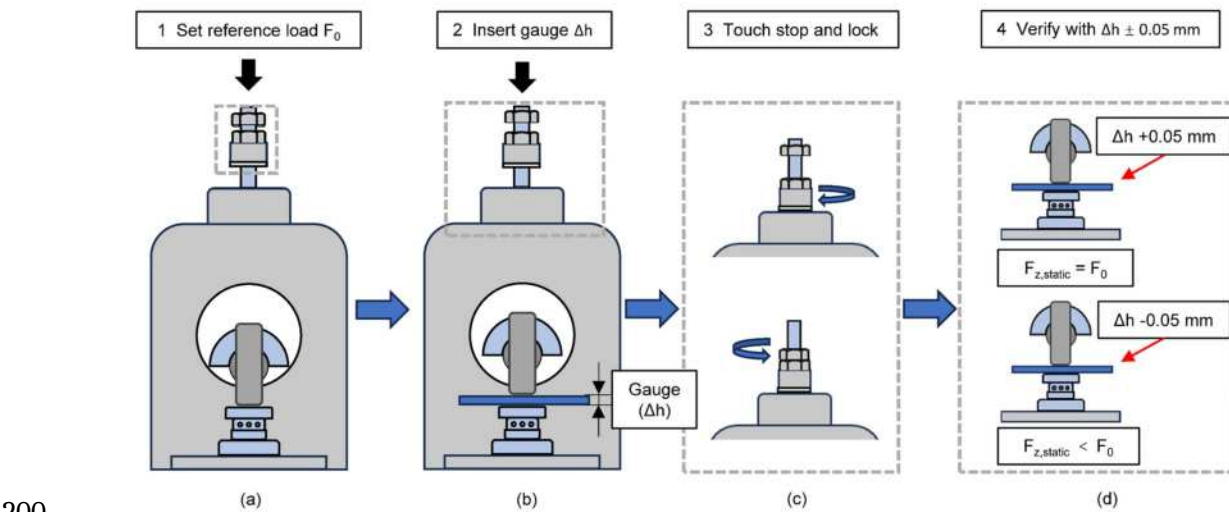


Fig. 2. Schematic illustration of the horn-anvil gap presetting procedure. (a) Set the reference load F_0 . (b) Insert the gauge Δh . (c) Touch the stop and lock the nuts. (d) Verify the setting with $\Delta h \pm 0.05$ mm gauges.

The main gap study was carried out at a fixed baseline of 20% generator amplitude, 0.25 s welding time, and 0.45 MPa nominal pneumatic pressure, while Δh was varied as 0, 0.5, 0.7, and 0.9 mm (Table 2). This baseline was selected from preliminary trials as a feasible operating window for isolating the effect of Δh under fixed oscillatory input and nominal load, rather than as a globally optimized parameter set. In addition, an auxiliary weld-time sweep at $\Delta h = 0.5$ mm was performed with the same amplitude and nominal pressure to probe the condition sensitivity of the monitoring-based interpretation presented later.

Table 2. Baseline ultrasonic welding parameters for the main gap study.

Parameter	Welding time, t (s)	Normal load, F_0 (N)	Generator amplitude (%)	Preset gap, Δh (mm)
Value	0.25	1380	20	0 / 0.5 / 0.7 / 0.9

2.3. Lap-shear strength testing

Single-lap shear tests were conducted on an MTS E43 system with a 10 kN load cell. Two 8 mm $\times$ 50 mm coupons welded with a 5 mm overlap produced a total specimen length of 95 mm with 15 mm grip sections and an effective gauge length of 65 mm (Fig. 1c). Tests were performed under displacement control at 0.8 mm min⁻¹. For each Δh condition, 20 specimens were tested, and the welding termination power was recorded for the same set of joints ($n = 20$ per Δh). Peak load, displacement at fracture, and failure mode were used to compare joint performance. The auxiliary weld-time series at $\Delta h = 0.5$ mm was tested using the same specimen geometry and loading procedure.

2.4. Microstructural characterization and nanoindentation

Fracture surfaces were examined by scanning electron microscopy (SEM, JSM-6380LV). Cross-sections were cut through the weld centre along the vibration direction (Fig. 1c), mechanically polished, electrolytically polished, and imaged in back-scattered electron mode to

225 reveal interface morphology, metallurgical bonding, and defects. Electron backscatter diffraction
226 (EBSD) was performed using an Oxford AZtecHKL system with a step size of 0.6 μm . Prior to
227 analysis, EBSD maps were cleaned by wild-spike removal and zero-solution cleanup. Grain size
228 was calculated as equivalent circle diameter using a minimum grain definition of five indexed
229 pixels. Kernel average misorientation (KAM), grain orientation spread (GOS), and orientation
230 distribution functions (ODFs) were used to quantify Δh -dependent microstructural evolution. The
231 0.6 μm EBSD mapping is used here primarily for comparative zone identification and
232 boundary/orientation statistics rather than as a full census of the finest sub-micrometre grains; the
233 smallest local refinement is additionally interpreted with support from the high-magnification
234 SEM/BSE observations.

235 Grain boundaries were classified using conventional misorientation thresholds: low-angle
236 grain boundaries (LAGBs) from 2° to 15° and high-angle grain boundaries (HAGBs) above 15° .
237 $\Sigma 3$ twin boundaries were identified using the Brandon criterion. For each Δh , representative
238 specimens were selected for cross-sectional SEM and EBSD analysis from the weld-centre region
239 where the largest deformation and highest temperature are expected. Because the $\Delta h = 0.5$ mm
240 condition exhibited the most pronounced recrystallization-related features, EBSD mapping was
241 repeated on an additional same-batch specimen at this condition (verification $n = 2$), and the same
242 characteristic seam features were reproduced. Multiple cross-sections were also inspected to
243 confirm that the observed interfacial morphologies and zone definitions were representative. The
244 monitoring values later discussed together with the cross-sectional structure were taken from the
245 same specimen subsequently used for SEM/BSE and EBSD analysis, enabling direct process-
246 structure correlation on the specimen level.

247 To further probe local micromechanical heterogeneity around the interface, selected-line
 248 nanoindentation was analysed across the weld-centre region. For each preset gap, a common 13-
 249 point line (about $180\ \mu\text{m}$ total span, about $15\ \mu\text{m}$ point spacing) was extracted from the indentation
 250 map after registration with the corresponding SEM cross-section, with the line centre aligned to the
 251 identified seam-centre position. Where more than one indentation dataset was available, the line
 252 with the clearest seam-centre registration was used for the cross-condition comparison; for $\Delta h =$
 253 $0.9\ \text{mm}$ this corresponded to the second dataset. All selected points were acquired at a common
 254 maximum load of about $80\ \text{mN}$. Hardness H and reduced modulus E_r were used as the primary
 255 comparison metrics, while maximum indentation depth h_{max} and contact stiffness S were retained
 256 as auxiliary indicators. The selected-line data were evaluated both as position-resolved profiles and
 257 as distributions of all raw points, so that average level and spatial dispersion could be compared
 258 together.

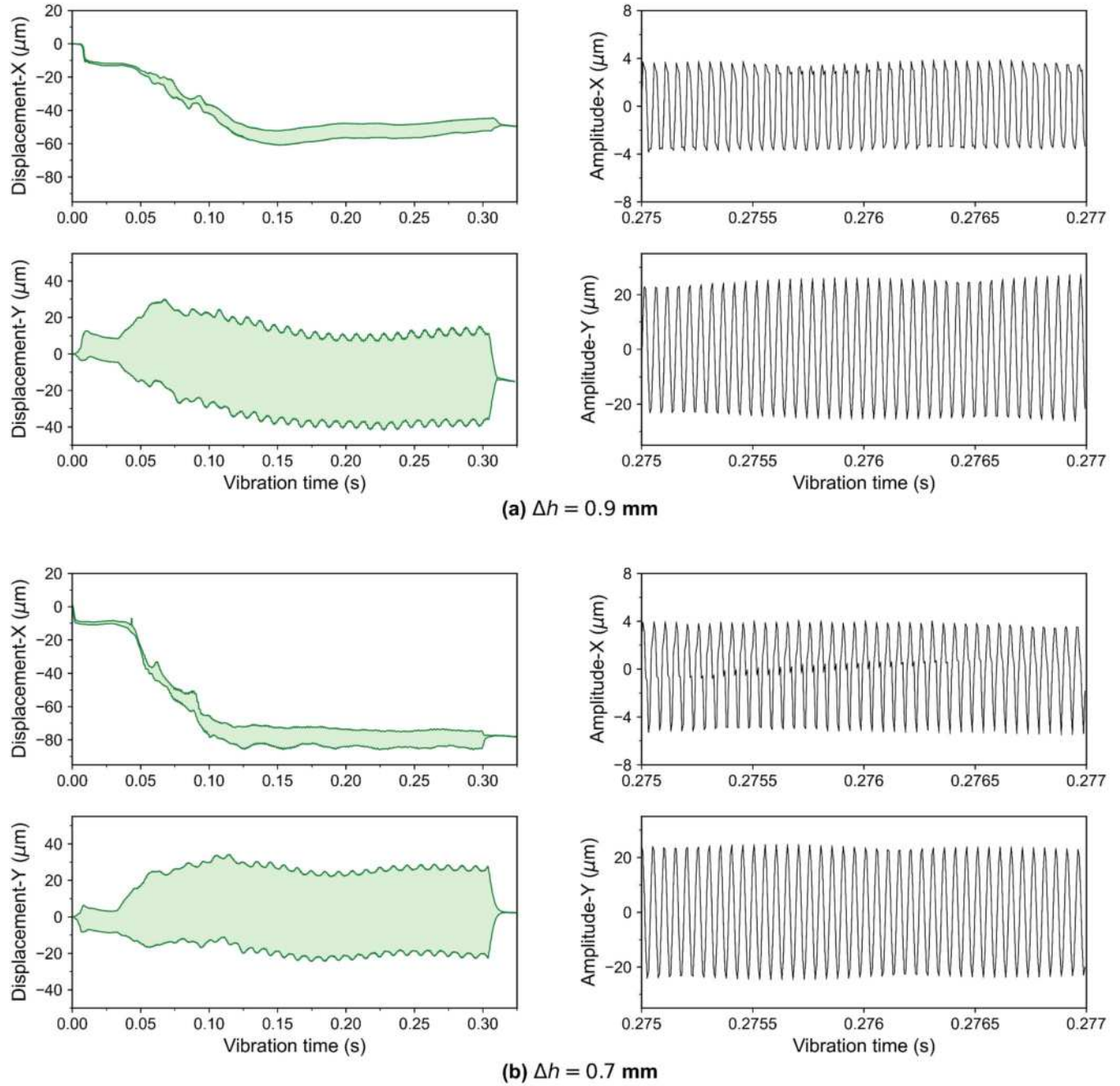
259 3. Results and analysis

260 3.1 Welding process monitoring data

261 3.1.1 Horn amplitude under different preset gaps Δh

262 Fig. 3 shows the in-process horn displacement histories in the sensor X and Y directions,
 263 corresponding respectively to specimen RD and TD, for $\Delta h = 0.9, 0.7, 0.5$ and $0\ \text{mm}$. The full raw
 264 displacement traces are shown together with their upper and lower envelopes, while the enlarged
 265 panels show zero-centred $2\ \text{ms}$ raw waveforms to highlight the local oscillatory amplitudes. At the
 266 start of vibration, both components exhibit a short low-amplitude plateau due to the phase-locked
 267 loop and soft-start strategy of the power supply. After ramp-up, the amplitude in the main Y
 268 direction rapidly reaches a stable level for all Δh , indicating that the preset gap has negligible

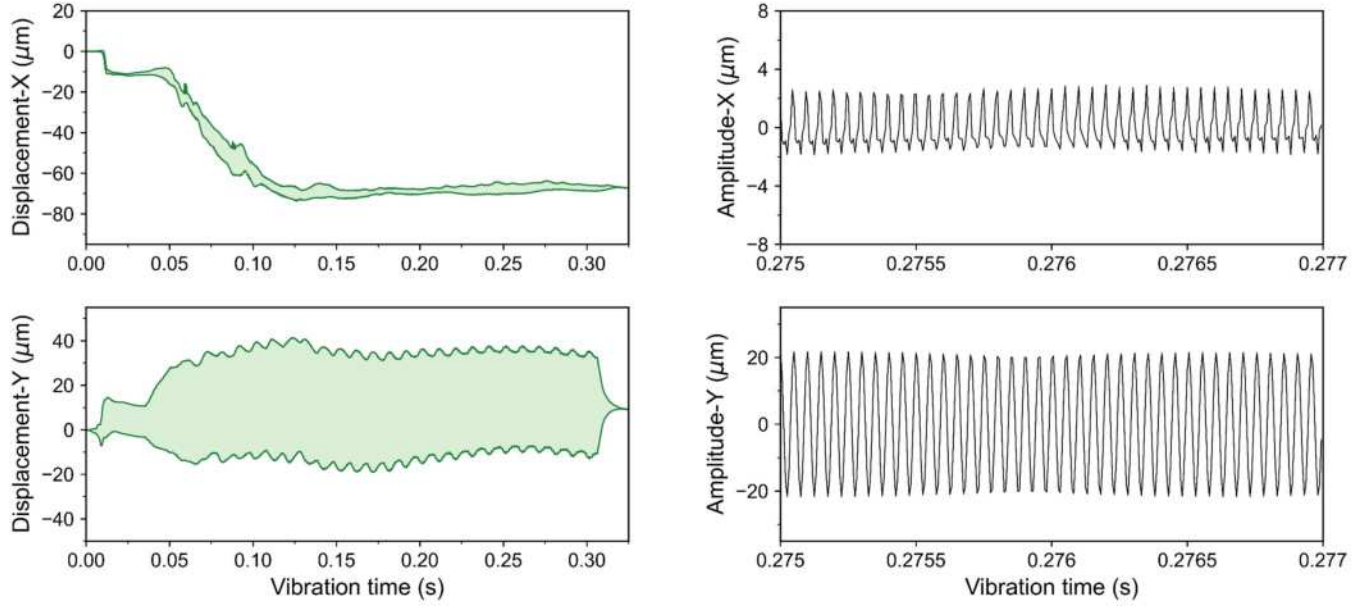
influence on the imposed displacement source. By contrast, the X-direction response remains much smaller and is interpreted here as a secondary off-axis coupled vibration of the horn/tooling system rather than as a second imposed vibration direction or a direct measure of interfacial slip.



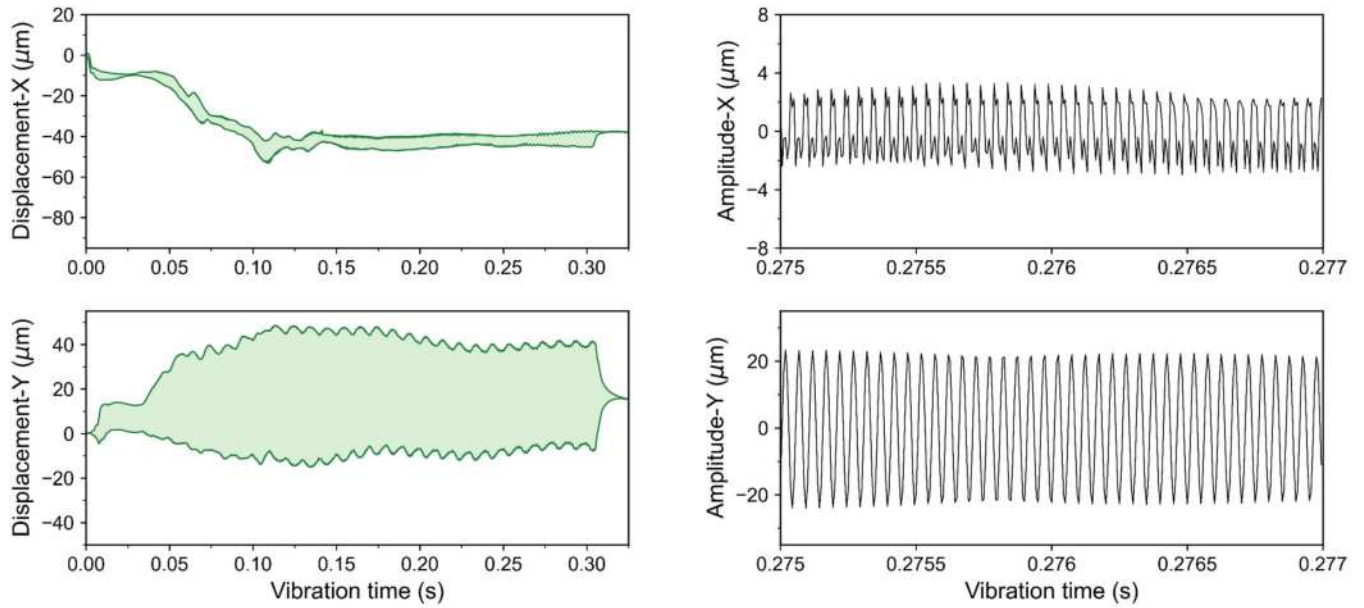
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Fig. 3. In-process horn displacement and local oscillatory amplitude in the sensor X and Y directions under different preset horn-anvil gaps Δh : (a) $\Delta h = 0.9 \text{ mm}$; (b) $\Delta h = 0.7 \text{ mm}$; (c) $\Delta h = 0.5 \text{ mm}$; (d) $\Delta h = 0 \text{ mm}$. The left panels show the full raw displacement histories with upper

276 and lower envelopes extracted using a 0.5 ms moving window with a 0.25 ms step. The right panels
 277 show zero-centred 2 ms raw waveforms to display the local oscillatory amplitudes. Sampling rate:
 278 160 kHz.



(c) $\Delta h = 0.5 \text{ mm}$



(d) $\Delta h = 0 \text{ mm}$

279

280 **Fig. 3.** (continued).

3.1.2 Welding loads and horn displacement under different preset gaps Δh

(1) Typical welding cycle and stage characteristics

Fig. 4 shows representative tri-axial force and horn-displacement histories for $\Delta h = 0.9$ mm.

The cycle comprises horn approach, clamping/pre-compression, ultrasonic vibration, hold, and horn retraction. During vibration, $u_{horn,z}$ decreases quasi-statically as the stack collapses, while F_x , F_y , and F_z oscillate strongly.

Within the ultrasonic stage, the force response is anisotropic. F_y mainly reflects the machine-side reaction to vibration-direction shearing, F_z the normal reaction, and F_x an off-axis response generated by three-dimensional knurled contact, lateral material flow, and structural coupling. F_x – F_y – F_z are therefore used as comparative machine-side descriptors of evolving contact stiffness and constraint.

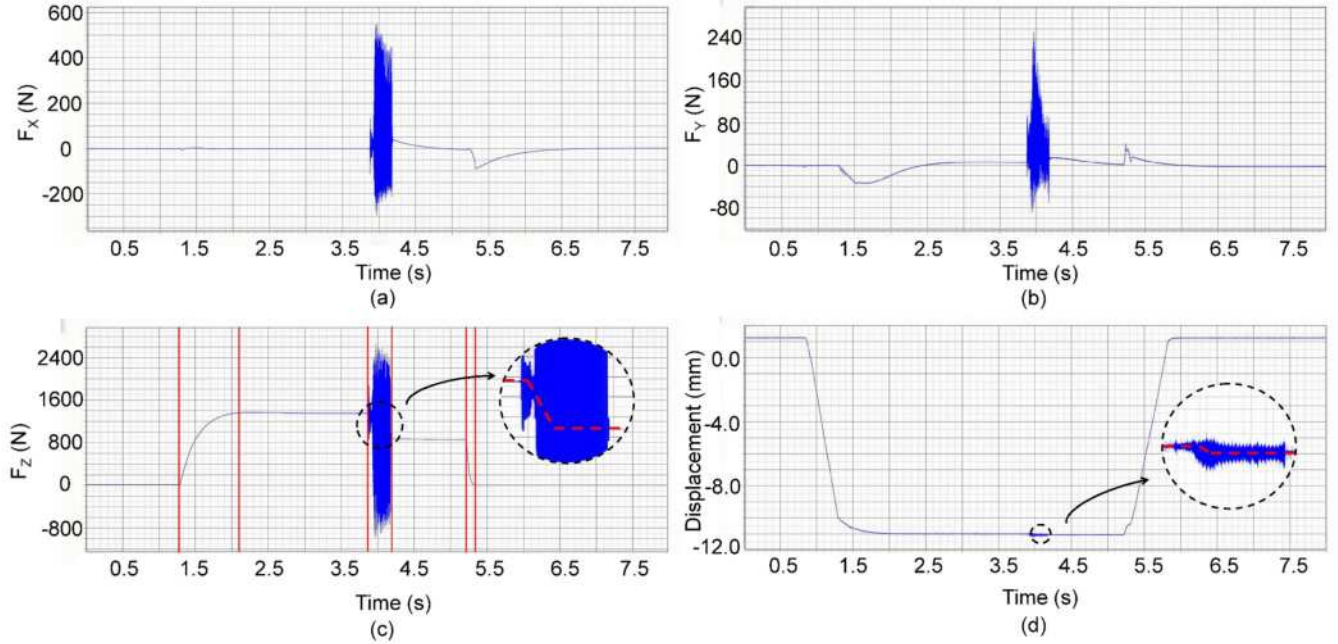


Fig. 4. Typical time histories of tri-axial forces and horn displacement in the Z direction over the whole welding cycle for $\Delta h = 0.9$ mm. Insets provide enlarged views of the signals during the ultrasonic stage together with the corresponding red dashed trend line used to reveal the local signal evolution.

(2) Definition of the effective ultrasonic loading window

To exclude start-up ramping and post-stop ringing, an effective ultrasonic loading window was defined by tracing 0.25 s backward from the disappearance of the high-frequency component in $u_{horn,z}$, as marked by the red dashed vertical lines in Fig. 5. For direct comparison among different Δh conditions, the time axis in Fig. 5 is referenced to the start of the refined effective ultrasonic loading window; therefore, $t=0$ and $t=0.25$ s denote the start and end of this window, respectively. Within this window, the force envelopes show a clear rise–peak–decay sequence, allowing consistent extraction of the stage durations t_1 and t_2 , together with the quasi-static horn approach, across all Δh conditions.

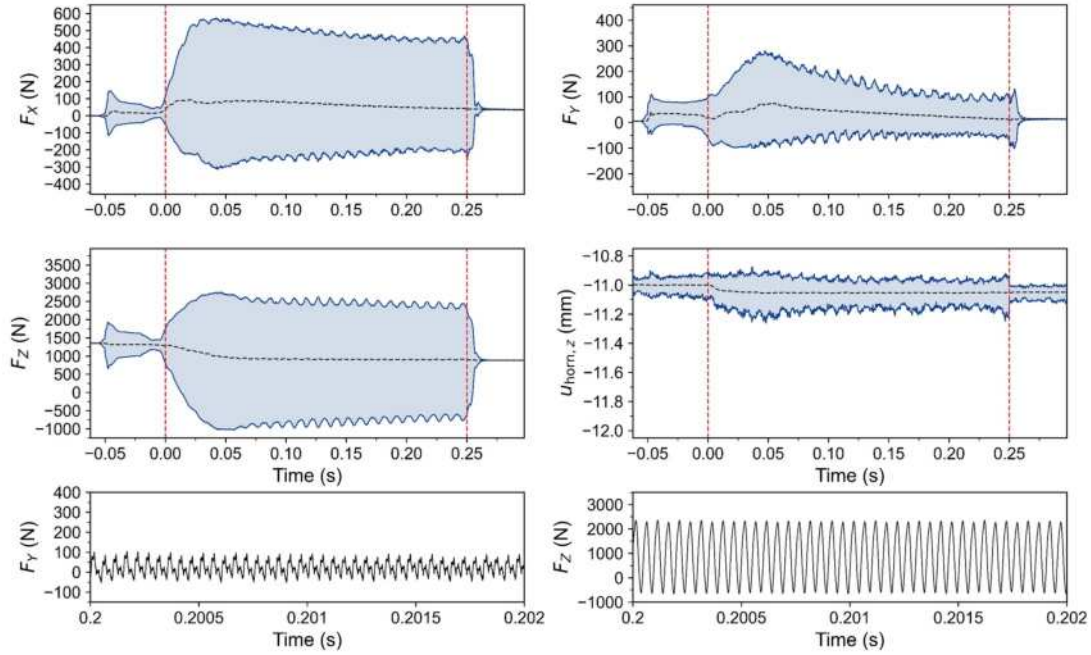
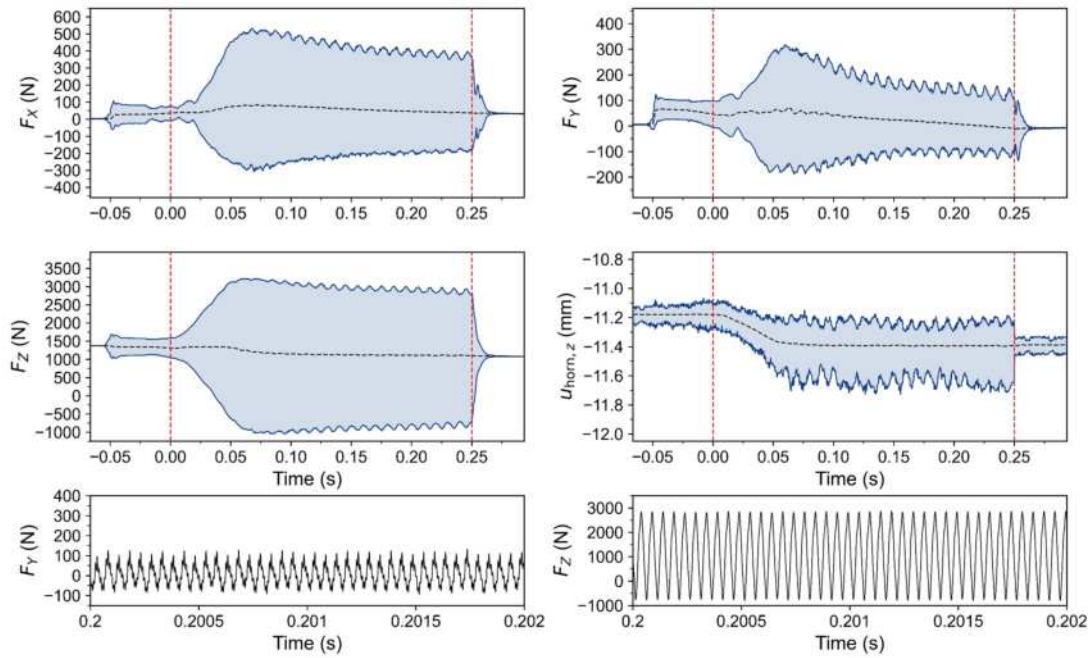
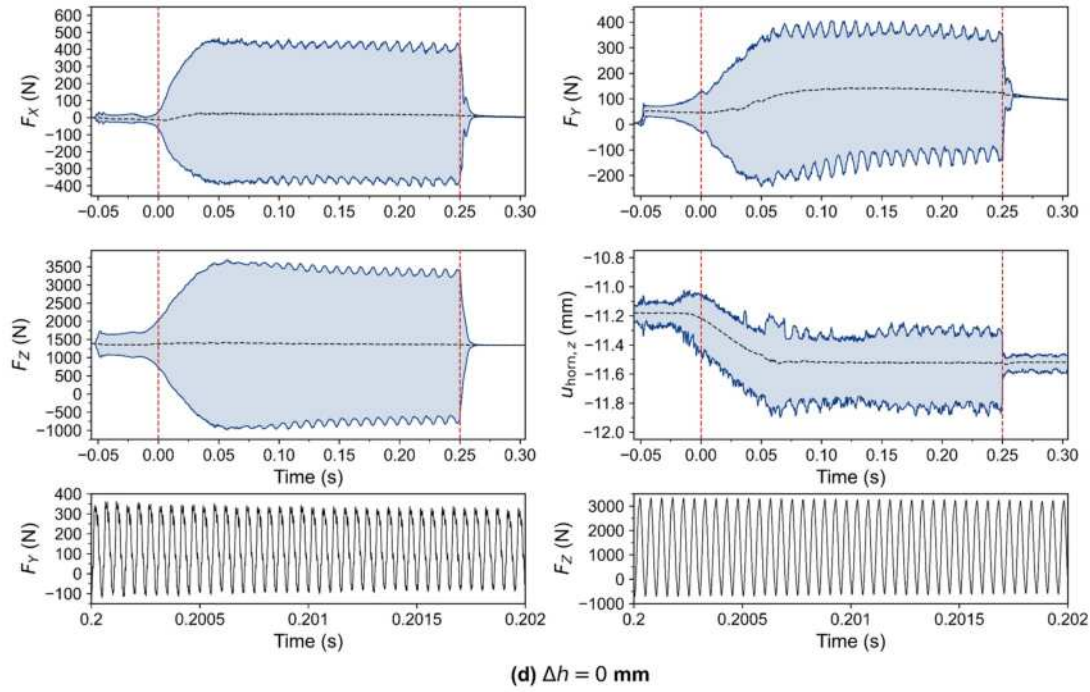
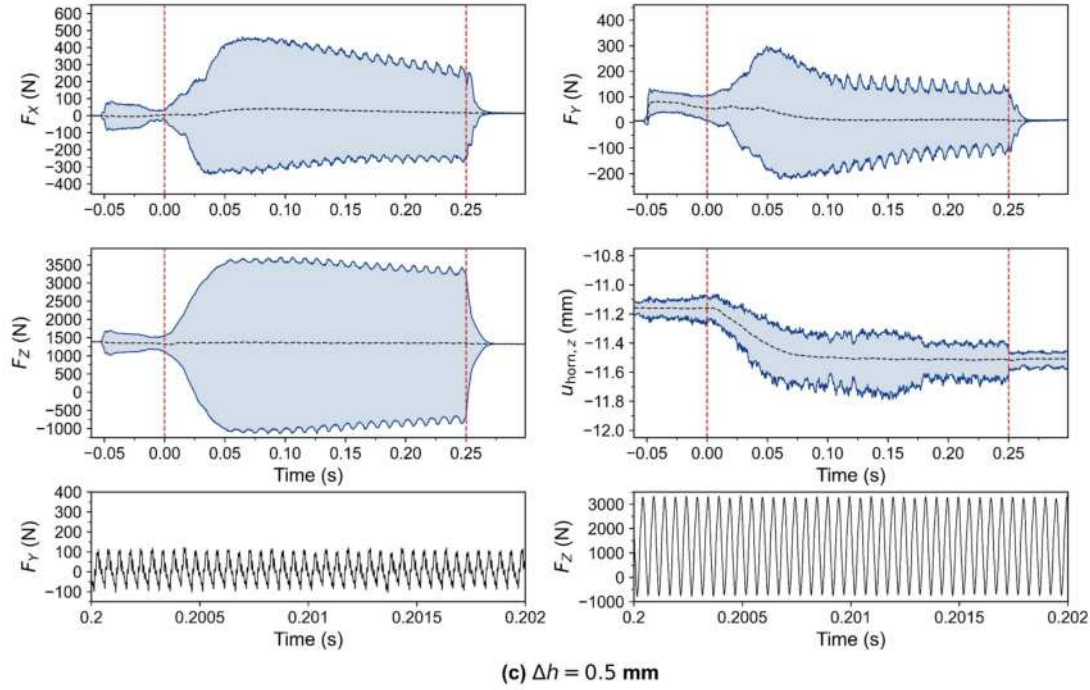
(a) $\Delta h = 0.9$ mm(b) $\Delta h = 0.7$ mm

Fig. 5. Enlarged in-process monitoring data of tri-axial forces F_x , F_y and F_z and horn displacement in the Z direction ($u_{horn,z}$) during the ultrasonic stage under different preset gaps Δh : (a) $\Delta h = 0.9$ mm; (b) $\Delta h = 0.7$ mm; (c) $\Delta h = 0.5$ mm; (d) $\Delta h = 0$ mm. The semi-transparent traces show the full raw records, and the upper/lower envelopes and dashed moving-mean centrelines

312 were extracted using a 0.5 ms moving window with a 0.25 ms step. Red dashed vertical lines mark
 313 the refined effective ultrasonic loading window. The lower panels show representative 2 ms raw
 314 F_Y and F_Z waveforms. Sampling rate: 1 MHz.



315

316 **Fig. 5. (continued).**

(3) Basic characteristics of the tri-axial dynamic forces

Within the effective window, each force component was separated into a slowly varying mean and a high-frequency envelope. The oscillatory loading is characterized here by the half peak-to-peak force envelope, $\tilde{F}_Y$, $\tilde{F}_X$, $\tilde{F}_Z$. For all conditions, the $\tilde{F}_Z$ is much larger than the $\tilde{F}_Y$, $\tilde{F}_X$, consistent with the fact that the horn–sheet–anvil system remains strongly normal-load dominated under the preset clamping force. Physically, the $\tilde{F}_Z$ reflects the periodic indentation/release response of the knurled horn–sheet contact and the high normal contact stiffness along the machine load path. By contrast, the $\tilde{F}_Y$ represents the measurable machine-side tangential response transmitted through the anvil/sensor path under the prevailing static normal pressure. It rises rapidly and then decays toward the late stage as the interface evolves from more sliding-dominated contact to a more constrained or partially stuck state. Accordingly, its late-stage level indicates whether appreciable frictional/plastic shearing is still being sustained, but it must always be interpreted together with the normal-load history rather than in isolation.

Quantitatively, $\tilde{F}_Z$ is on the order of 1800–2400 N, whereas the $\tilde{F}_Y$ is about 180–280 N. $\tilde{F}_X$ is nevertheless substantial and can reach about 370–440 N even though the imposed ultrasonic amplitude is much larger in Y than in X. This apparent mismatch shows that the measured X-force is not governed by the X-amplitude alone. In the Y direction, the normal load mainly affects the measured response by setting the interfacial shear resistance under the prescribed oscillatory motion through its effect on real contact, frictional traction and tangential stiffness. In the X direction, where no principal vibration is imposed, part of the large normal-load response can be redistributed into a non-cancelling off-axis reaction because the real knurled contact is discrete and asymmetric, with lateral material flow/folding, tooth-scale engagement differences, local stick-slip heterogeneity, and horn–sheet–anvil–fixture coupling. The Y channel should therefore be regarded

as the primary imposed-vibration direction, whereas the X channel is retained mainly as a diagnostic indicator of off-axis constraint, contact asymmetry and three-dimensional machine-side contact evolution, rather than as evidence of a second imposed vibration direction or a direct transverse interfacial shear stress.

(4) Comparison of characteristic quantities of monitoring data under different Δh

Table 3 shows that the stage durations t_1 and t_2 vary little with Δh , whereas collapse and late-stage load evolution change markedly. Final horn approach increases from 0.07 mm at $\Delta h = 0.9$ mm to 0.23 mm at 0.7 mm and then effectively saturates at 0.36 mm for $\Delta h \leq 0.5$ mm. Over the same range, static-force decay decreases strongly, while $\tilde{F}_{Y,max}$ changes only moderately. The most discriminating quantity is $\tilde{F}_{Y,min}$: it remains low for $\Delta h = 0.9$ –0.5 mm but rises sharply to 226.0 N at $\Delta h = 0$ mm, indicating persistent late-stage shearing once collapse is no longer effectively bounded. The representative values remain within the repeatability range of the monitored subset ($n = 10$ per Δh). $\tilde{F}_{Y,min}$ must nevertheless be interpreted together with collapse and static-force history, because a low value may also reflect premature unloading, as at $\Delta h = 0.9$ mm.

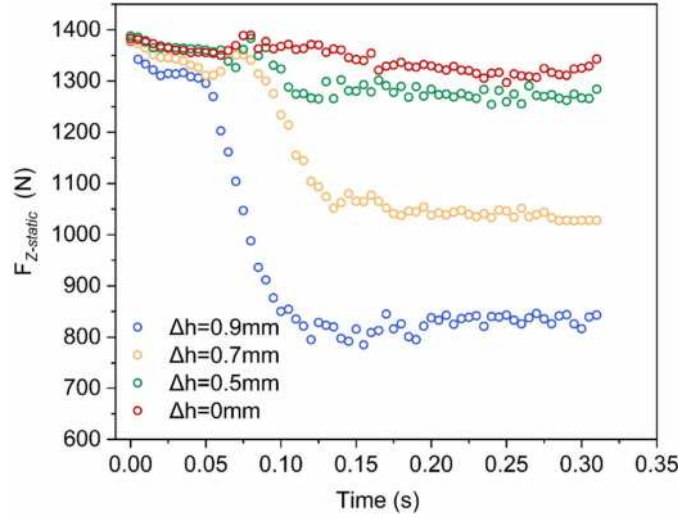
Table 3. Characteristic quantities extracted from in-process monitoring data within the effective ultrasonic window (representative values from the monitored subset; $n = 10$ per Δh).

$\Delta h(\text{m})$	$t_1(\text{s})$	$t_2(\text{s})$	$\tilde{F}_{X,max}(\text{N})$	$\tilde{F}_{Y,max}(\text{N})$	$\tilde{F}_{Z,max}(\text{N})$	$\tilde{F}_{X,min}(\text{N})$	$\tilde{F}_{Y,min}(\text{N})$	$\tilde{F}_{Z,min}(\text{N})$	$u_{horn,z}(\text{mm})$	$\Delta F_{Z,static}(\text{N})$
0.9	0.049	0.201	436.5	177.0	1813.0	322.5	72.5	1467.0	0.07	472
0.7	0.061	0.189	377.0	240.5	2026.5	241.0	82.0	1531.5	0.23	303
0.5	0.064	0.186	368.0	234.5	2329.0	242.0	80.0	1662.5	0.36	74
0	0.058	0.192	414.5	276.5	2362.0	371.0	226.0	1998.5	0.36	45

(5) Evolution of static normal force

Fig. 6 directly shows how Δh reshapes the static normal-force history. At large gaps the stop is engaged early and further deformation mainly unloads machine compliance, causing pronounced decay of $F_{Z,static}$. As Δh decreases, the onset of relaxation shifts later, and for $\Delta h \leq 0.5$

mm the static force remains close to F_0 through most of the weld. Thus Δh directly schedules the time-domain normal-pressure history under otherwise fixed conditions.

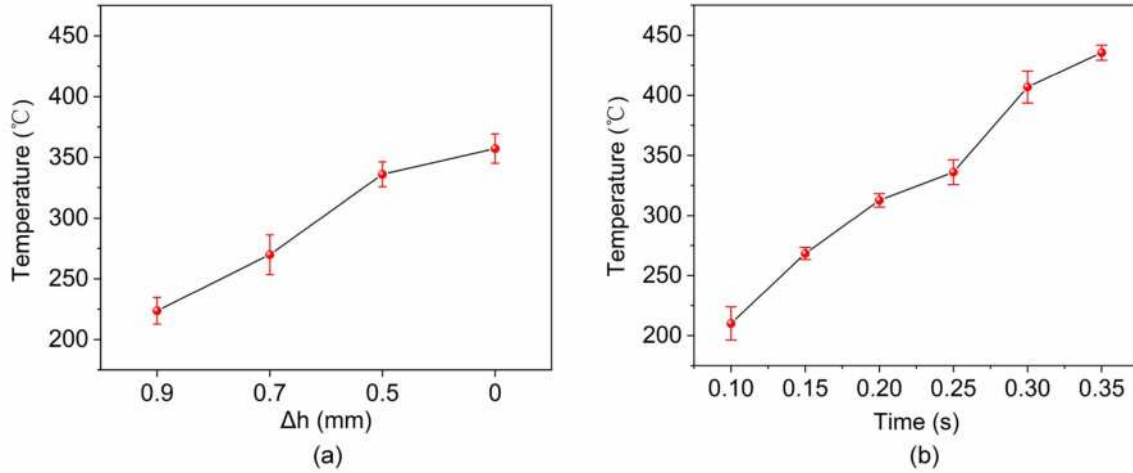


362

363 **Fig. 6.** Evolution of the static normal force $F_{Z,static}$ during ultrasonic vibration for different preset
364 horn-anvil gaps Δh .

365 3.1.3 Weld temperature and power response

366 Fig. 7a shows that decreasing the preset horn-anvil gap distance Δh increases the weld-centre
367 peak temperature measured near the faying interface. T_{peak} rises from 223.7 ± 10.9 °C at $\Delta h = 0.9$
368 mm to 357.2 ± 12.1 °C at $\Delta h = 0$ mm (mean ± 1 SD, $n = 3$). Notably, the weld-centre peak
369 temperature at $\Delta h = 0$ mm exceeds the reported Curie-temperature range of Invar, suggesting that
370 the Invar effect may be temporarily suppressed during welding. For $\Delta h = 0.5$ mm, Fig. 7b shows
371 that T_{peak} increases from 210.1 ± 13.8 °C at 0.10 s to 336.0 ± 10.3 °C at 0.25 s, followed by a rapid
372 rise to 406.9 ± 13.3 °C at 0.30 s and 435.5 ± 6.3 °C at 0.35 s. This trend also shows that the
373 mechanically robust intermediate-gap conditions are not those with the highest thermal input.



374

375 **Fig. 7.** Peak temperature measured near the faying interface (mean $\pm$ 1 SD, $n = 3$): (a) with different
 376 Δh ; (b) with different welding time ($\Delta h = 0.5$ mm).

377 Termination power at the end of welding (Fig. 8) increases with decreasing Δh . The grouped
 378 distributions show a clear upward shift in the median from $\Delta h = 0.9$ to 0.7 mm, followed by a
 379 further but more gradual increase at 0.5 and 0 mm. For $\Delta h = 0$ mm, both the interquartile range and
 380 the overall spread are the largest, implying that the equivalent mechanical impedance of the horn-
 381 stack–anvil system becomes more sensitive to local variations when the process is driven into the
 382 zero-gap over-processed regime.

383 In ultrasonic welding systems, generator power reflects the instantaneous mechanical
 384 impedance of the horn-stack–anvil system, which evolves as contact stiffness, frictional conditions
 385 and local temperature change. Therefore, the observed increase in termination power with
 386 decreasing Δh is consistent with a stiffer, more strongly coupled contact state under higher
 387 sustained normal force and larger collapse. The large scatter at $\Delta h = 0$ mm further indicates that
 388 small variations in surface condition or local deformation can significantly perturb impedance in
 389 the zero-gap condition, which supports using power jointly with force/displacement monitoring for
 390 process stability assessment.

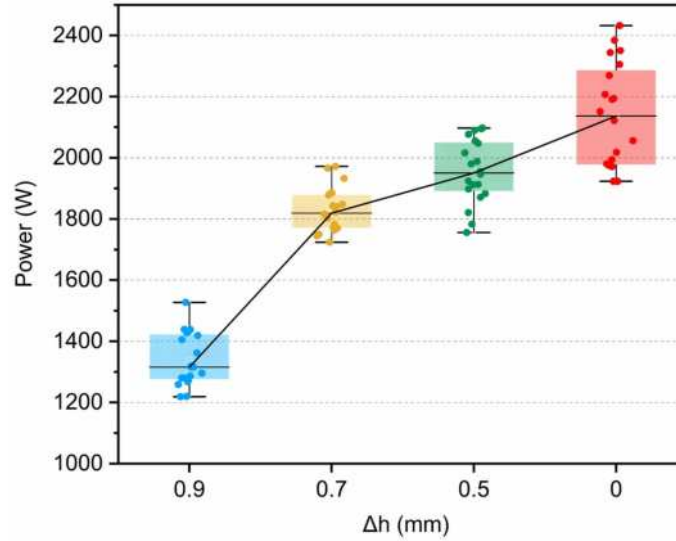


Fig. 8. Ultrasonic termination power grouped by preset horn-anvil gap Δh ($n = 20$ per gap). Boxes show median and IQR; whiskers show $1.5 \times \text{IQR}$; dots show individual specimens; the solid line connects the group medians.

3.2 Effect of preset gap Δh on lap-shear strength of the joints

Twenty lap-shear specimens were tested for each Δh . Fig. 9a shows that mechanical performance depends strongly on Δh . Intermediate gaps ($\Delta h = 0.5\text{--}0.7$ mm) deliver the highest combination of peak load and fracture displacement, whereas both the large-gap and zero-gap conditions are inferior. Within the intermediate window, $\Delta h = 0.7$ mm gives the highest mean displacement and $\Delta h = 0.5$ mm a very similar mean peak load. The differences in mean peak load and fracture displacement between $\Delta h = 0.5$ and 0.7 mm were small relative to the within-condition scatter, so these two conditions are interpreted here as neighbouring states within a common robust high-performance window rather than as statistically well-separated optima.

Three fracture modes were identified: interface fracture, transverse through-thickness (TTT) fracture in the thinned plastic-flow zone, and base-metal fracture. Fig. 9b summarises the mode counts ($n = 20$ per Δh). Interface fracture dominates at $\Delta h = 0.9$ mm (17/20), disappears at 0.7 mm (0/20), reappears slightly at 0.5 mm (2/20), and increases again at 0 mm (4/20). The remaining

failures at 0.7 and 0.5 mm are split between TTT and base-metal modes, whereas at 0 mm TTT becomes the dominant non-interface failure. Thus, $\Delta h = 0.5\text{--}0.7$ mm minimises interface-related failure, and $\Delta h = 0.7$ mm is practically safer because no interface fracture was observed in the present dataset.

These mode changes clarify the role of the processing window. Interface fracture indicates insufficient interfacial bonding or residual oxide, TTT fracture indicates that the interface has become stronger than the adjacent thinned zone, and base-metal fracture indicates that the welded region can sustain plastic deformation without catastrophic interfacial failure. The renewed interface-related failures at $\Delta h = 0$ mm therefore imply damage accumulation or heterogeneity under the zero-gap condition rather than simple under-bonding.

Fig. 9c further relates fracture mode to the load–displacement response. Interface-fracture joints cluster at lower loads and small displacements, base-metal fractures occupy the high-load/high-displacement region, and TTT fractures span an intermediate range. Notably, the TTT displacement range is narrowest at $\Delta h = 0$ mm, whereas at $\Delta h = 0.7$ and 0.5 mm it extends to larger displacements with maximum loads comparable to base-metal fracture.

At $\Delta h = 0.5$ mm, increasing welding time from 0.10 to 0.25 s raises peak load and shifts the dominant failure mode from TTT toward base-metal fracture (Fig. 9d). When welding time is further increased to 0.30–0.35 s, both load and displacement decrease and interface-related or TTT failure reappears, consistent with the strong temperature rise in Fig. 7b.

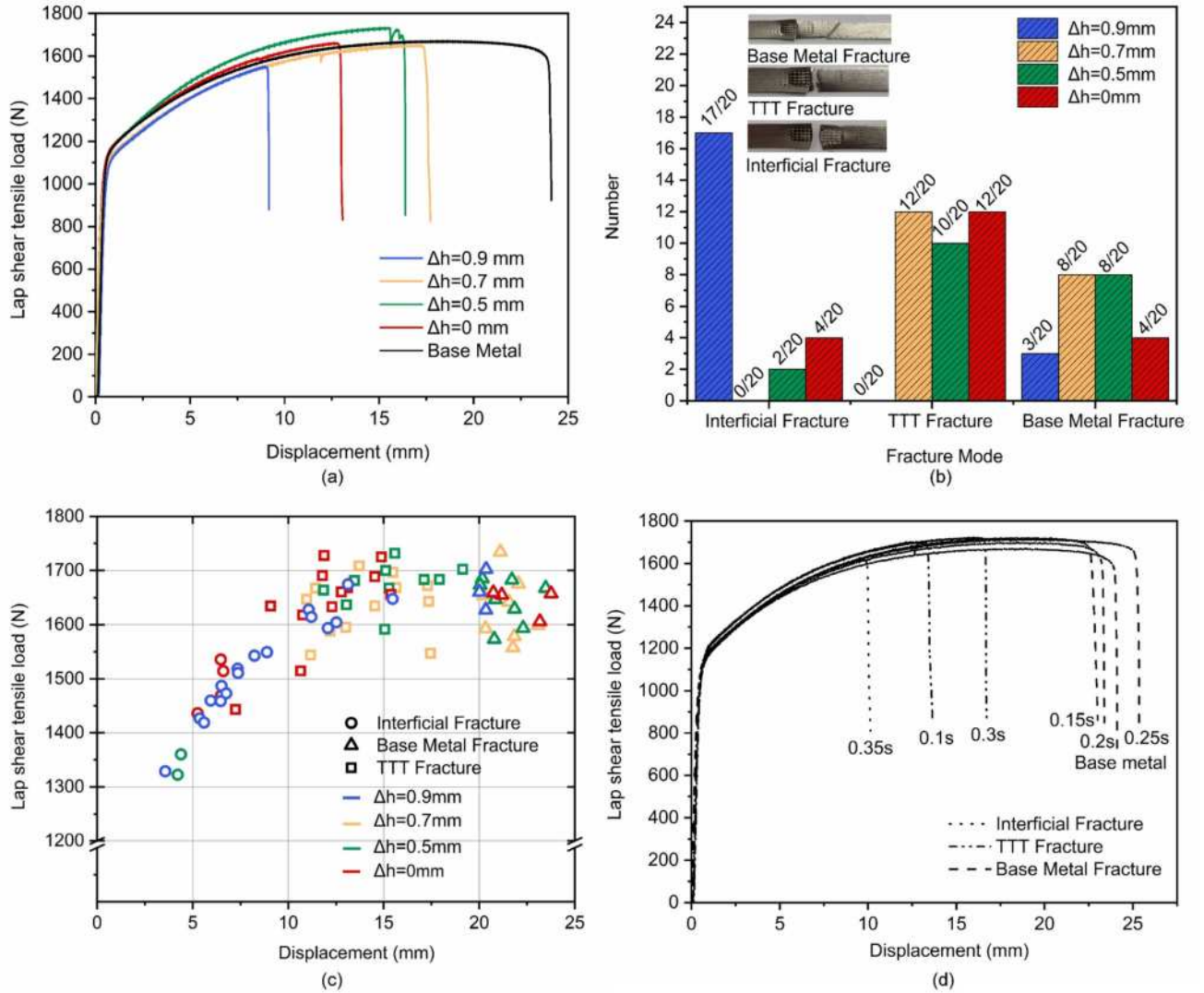
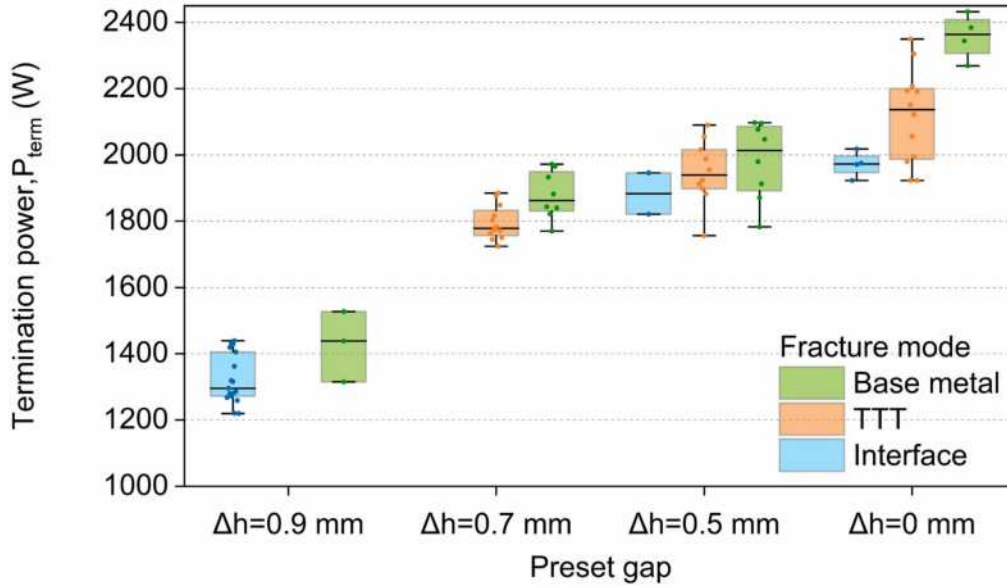


Fig. 9. Lap-shear performance and fracture behaviour of Invar-Invar ultrasonic welded joints under different preset gaps Δh . (a) Representative lap-shear load-displacement curves for different Δh . (b) Fracture-mode distribution ($n = 20$ per Δh ; numbers above bars show x/20 specimens). (c) Scatter plot of fracture load versus fracture displacement of 20 tested joints for each preset gap. (d) Lap-shear load-displacement curves of joints welded with different welding times at $\Delta h = 0.5$ mm.

Fig. 10 shows that, for a given Δh , interface fractures cluster at the low-power end of the termination-power distribution, base-metal fractures at the high-power end, and TTT fractures usually in between. However, the overlap is substantial, especially between TTT and base-metal fractures, so power alone is not a sufficient quality criterion. This overlap is precisely why P_{term} is

used here only as an auxiliary impedance signal under fixed Δh and is subsequently combined with collapse and force-retention descriptors in Section 4.2.3.



439

Fig. 10. Termination power grouped by fracture mode within each preset gap ($n = 20$ per gap). Boxes show median and IQR; whiskers show $1.5 \times \text{IQR}$; dots show individual specimens.

441

3.3 Fracture modes and fractography

442

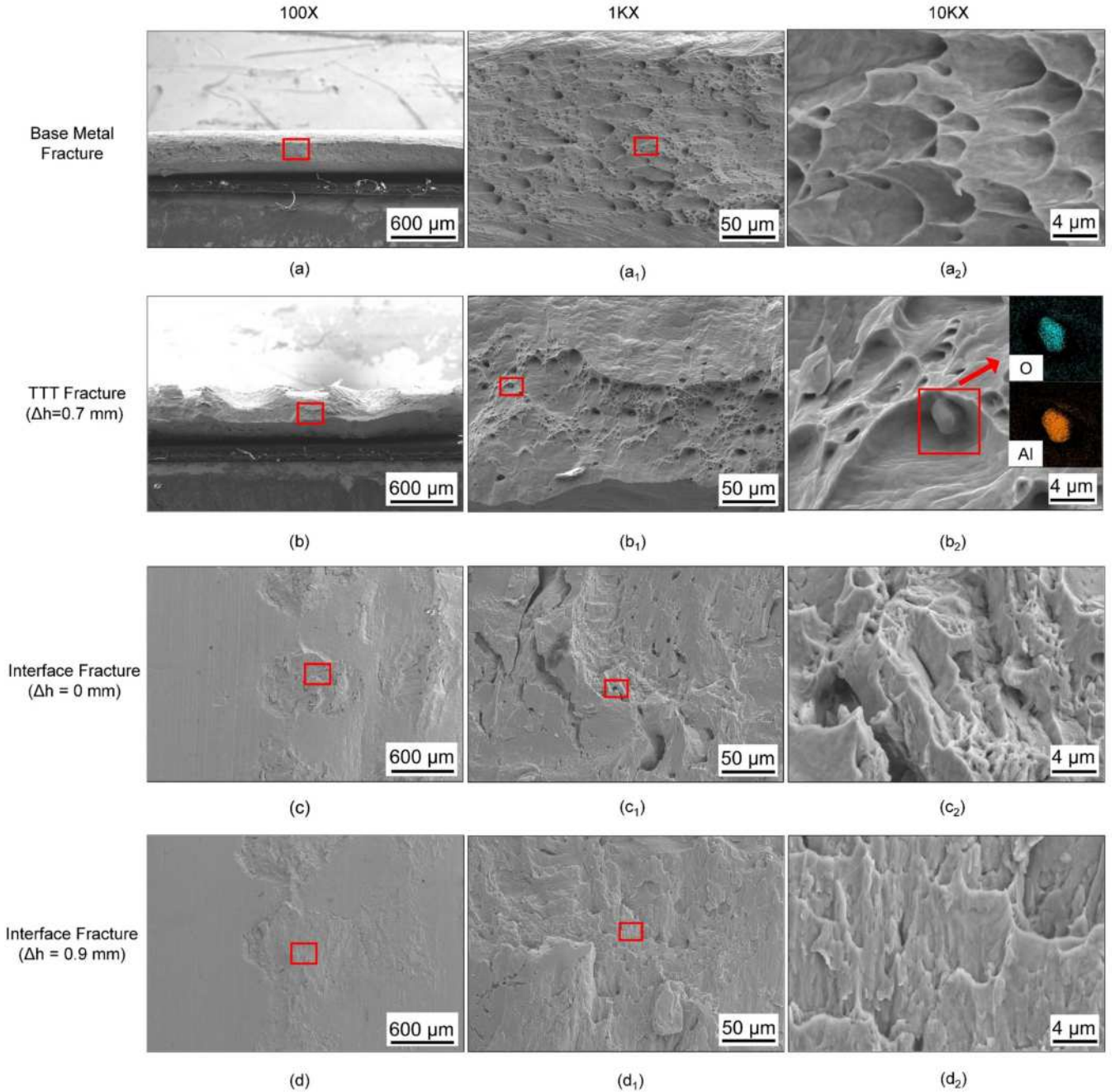
Fig. 11 compares representative SEM fracture morphologies for base-metal fracture, TTT fracture and interface fracture under different Δh . Base-metal fractures exhibit an inclined fracture plane ($\sim 45^\circ$ to the loading direction) with pronounced necking away from the weld and surfaces covered by deep, equiaxed dimples, consistent with micro-void nucleation and coalescence in an austenitic Invar matrix^[21,22]. TTT fractures (typical at $\Delta h = 0.7$ mm) remain dimple-dominated but with slightly smaller and shallower dimples; high-magnification images and EDS mapping, as highlighted in Fig. 11b₂, reveal O- and Al-rich particles at some dimple centres, suggesting void nucleation at fragmented oxide particles^[21,22].

450

451 Cross-sectional inspection confirms that TTT cracks initiate and propagate within the severe-
 452 thinning band adjacent to the weld, where strain concentrates during lap-shear loading. The
 453 reduced dimple size compared with the base metal is consistent with higher local strength and
 454 lower void growth capacity, which can arise from grain refinement, substructure formation and
 455 residual oxide/particle fragmentation in the plastic-flow zone. Notably, no large cleavage facets are
 456 observed in the base-metal or TTT fractures, supporting the interpretation that the dominant failure
 457 mechanism remains ductile micro-void coalescence in Invar 36 for well-bonded joints.

458 In contrast, interface-proximal failures at $\Delta h = 0$ mm show mixed ductile–brittle morphology:
 459 only limited areas contain small dimples, whereas quasi-cleavage facets, river patterns and tearing
 460 ridges occupy large regions. In fcc Invar these features are better described as quasi-cleavage rather
 461 than true cleavage, indicating accelerated crack propagation along locally hardened or embrittled
 462 bands while still accompanied by plasticity ^[21,22]. For the large-gap condition $\Delta h = 0.9$ mm, fracture
 463 surfaces are nearly planar and smooth at low magnification, with broad flat facets and pronounced
 464 river patterns and very few dimples, consistent with brittle quasi-cleavage-type interface fracture
 465 along weakly bonded regions or residual oxides ^[23,24]. Overall, intermediate gaps $\Delta h = 0.5$ – 0.7 mm
 466 promote fully dimpled TTT/base-metal fracture, whereas large- or zero-gap conditions promote
 467 brittle or mixed interface-proximal failures.

468 These contrasts indicate that zero-gap failure is associated with damage-localized over-
 469 processing and spatially heterogeneous seam evolution. By comparison, large-gap failure reflects
 470 incomplete bonding and limited interlocking rather than late-stage damage accumulation.



471

472 **Fig. 11.** SEM fracture morphologies of Invar-Invar ultrasonic welded joints produced with
 473 different preset horn-anvil gaps Δh . (a–a₂) Base-metal fracture; (b–b₂) transverse through-
 474 thickness (TTT) fracture at $\Delta h = 0.7$ mm; (c–c₂) interface-proximal mixed fracture at $\Delta h = 0$ mm;
 475 (d–d₂) interface fracture at $\Delta h = 0.9$ mm.

3.4 Interfacial microstructure under different preset gaps Δh

Fig. 12 shows cross-sectional SEM images of the weld region at different magnifications. The preset gap controls the minimum remaining thickness, the continuity and morphology of the interface, local flow patterns and the development of refined bands in the interfacial region. Unless otherwise noted, the cross-sectional microstructural results presented below were obtained from the same representative monitored specimens listed in Table 3.

At intermediate and small gaps, the interfacial region can be described by a fine-grained weld seam (WS) embedded between transition zones, consistent with the EBSD-based zoning in Section 3.5. The morphological continuity of the WS and the degree of mechanical interlocking are closely aligned with the observed fracture modes: discontinuous WS and retained interfacial gaps correlate with interface fracture, whereas a continuous WS with extensive intrusion and folding correlates with TTT/base-metal failure.

3.4.1 Weld thickness and its relation to Δh

As Δh decreases from 0.9 to 0.7, 0.5 and 0 mm, the minimum weld thickness generally decreases from about 623 to 554, 510 and 526 μm , respectively. In the present study, this metric is used only as a comparative descriptor of local thinning severity on the analysed 2D centreline cross-section under different Δh conditions. The apparent reversal between $\Delta h = 0.5$ and 0 mm mainly reflects specimen-to-specimen scatter (edge loss during preparation and slight horn-tooth wear); horn-displacement monitoring shows that total collapse is almost identical for these two conditions, indicating that additional compression is limited under the present welding settings. It should therefore not be interpreted as a direct measure of the three-dimensional average load-bearing area of the lap region, but only as a centreline descriptor of local thinning severity for cross-condition comparison.

3.4.2 $\Delta h = 0.9$ mm: discontinuous interface and weak mechanical interlocking

At $\Delta h = 0.9$ mm, a distinct, nearly planar interface remains between the sheets and bonded regions are discontinuous. Obvious gaps persist in peak–peak, valley–valley and peak–valley regions of the tool-tooth pattern, and the effective nugget is narrow. High-magnification images show a mostly straight, single-layer interface with only a thin refined-grain band on both sides; occasional wedge-like intrusions occur, but the length scale of grain refinement and mechanical interlocking is limited.

3.4.3 $\Delta h = 0.7$ mm: well-developed bonding, strong mechanical interlocking and extensive grain refinement

Reducing Δh from 0.9 to 0.7 mm produces a continuous, wavy interface with only narrow isolated gaps in misaligned regions and a fully closed peak–peak region. The interface is embedded in a wide refined band. Equiaxed fine grains nucleated along the original interface and the serrated, zig-zag interface geometry indicate dynamic recrystallisation. Grains from the two sheets mutually intrude and undergo intense tangential shear, forming pronounced mechanical interlocking; locally stacked interface layers suggest successive collapse and secondary shearing.

3.4.4 $\Delta h = 0.5$ mm: narrow refined band and more localised shear

At $\Delta h = 0.5$ mm, the interface becomes difficult to distinguish at low magnification, and high magnification reveals a narrow band of strongly refined, elongated grains embedded in a coarser matrix. Compared with $\Delta h = 0.7$ mm, the refined zone is narrower but more concentrated, indicating that plastic deformation and grain refinement become more localised into a shear-band-like region while the surrounding material experiences less extensive recrystallisation.

520 3.4.5 $\Delta h = 0$ mm: layered ultrafine-grain shear band under continued downward stroke

521 For $\Delta h = 0$ mm, the horn continues to move downward throughout the 0.25 s vibration stage.
522 The interface evolves into a very thin streamline-shaped overlapping band populated by sub-
523 micrometre grains, often accompanied by small voids and inclusions. This layered ultrafine-grain
524 band is consistent with repeated shear and compression under persistently high normal force.
525 Compared with the pronounced interlocking seen at intermediate gaps, the interface is
526 progressively flattened and regenerated into a multi-layer band associated with strong strain
527 localisation and higher damage susceptibility.

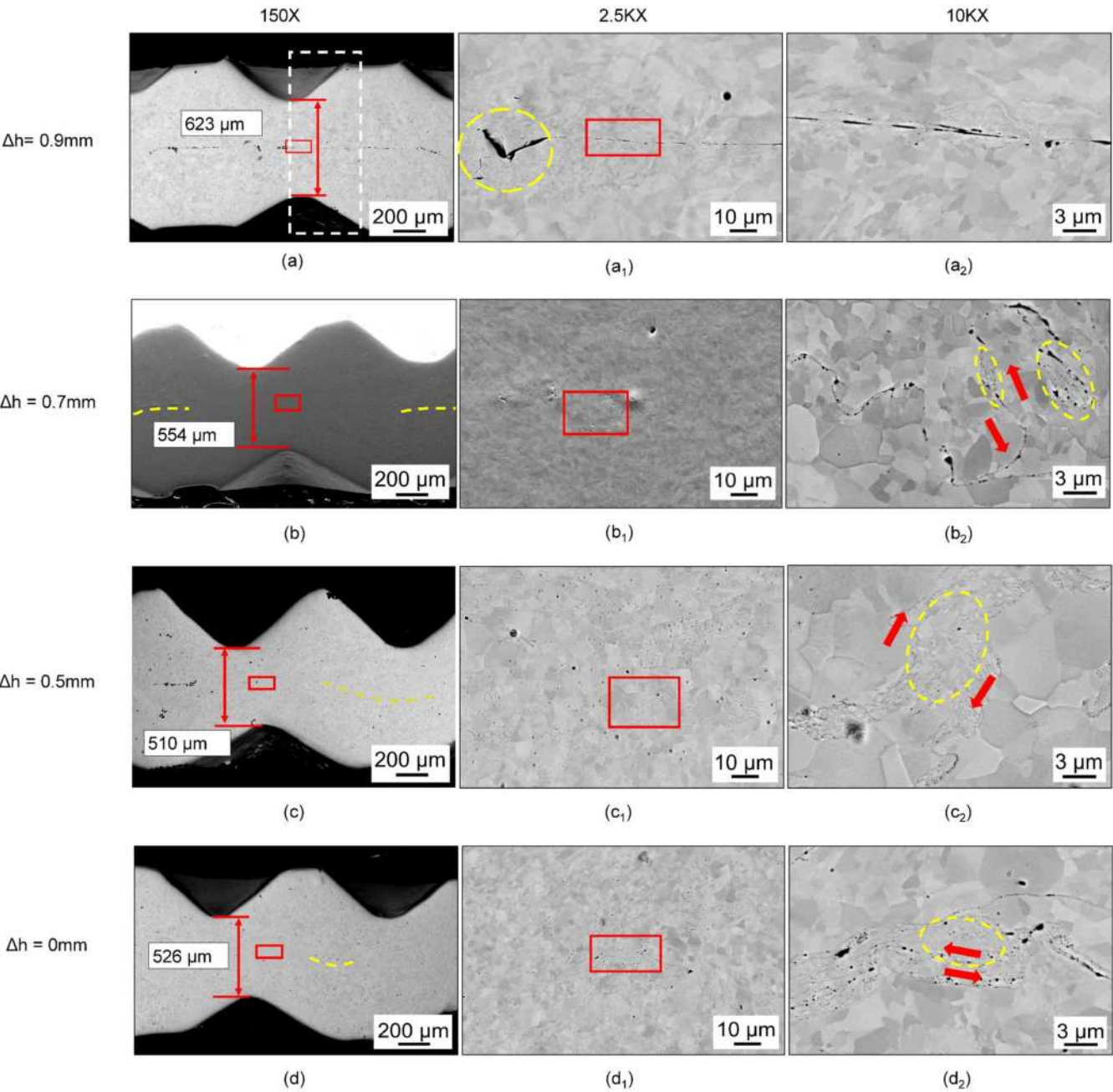


Fig. 12. Macrographic interface morphology and micrographic interface microstructure of welded joint cross-sections under different Δh conditions.

531 3.5 Crystallographic characterization and nanoindentation response of the weld interface

532 3.5.1 EBSD characteristics of the weld interface region

533 EBSD was performed on representative ND–TD sections of the base metal and joints welded
534 at $\Delta h = 0.9, 0.7, 0.5$, and 0 mm (Fig. 13). KAM, GOS, grain size, and grain-boundary statistics
535 were used to characterize the interfacial microstructure. Four characteristic regions were identified
536 through the thickness: the surface-affected zone (SAZ), coarse-grain transition zone (CGTZ), fine-
537 grain transition zone (FGTZ), and weld seam (WS). Here, the interfacial region refers to the WS
538 together with the adjacent FGTZ. The base metal consists of homogeneous equiaxed grains with an
539 average size of about $15.8\text{ }\mu\text{m}$, very low internal misorientation (median KAM $\approx 0.21^\circ$), and a high
540 fraction of $\Sigma 3$ annealing twins (about 55%), which together define the reference state for the Δh -
541 dependent interfacial changes described below.

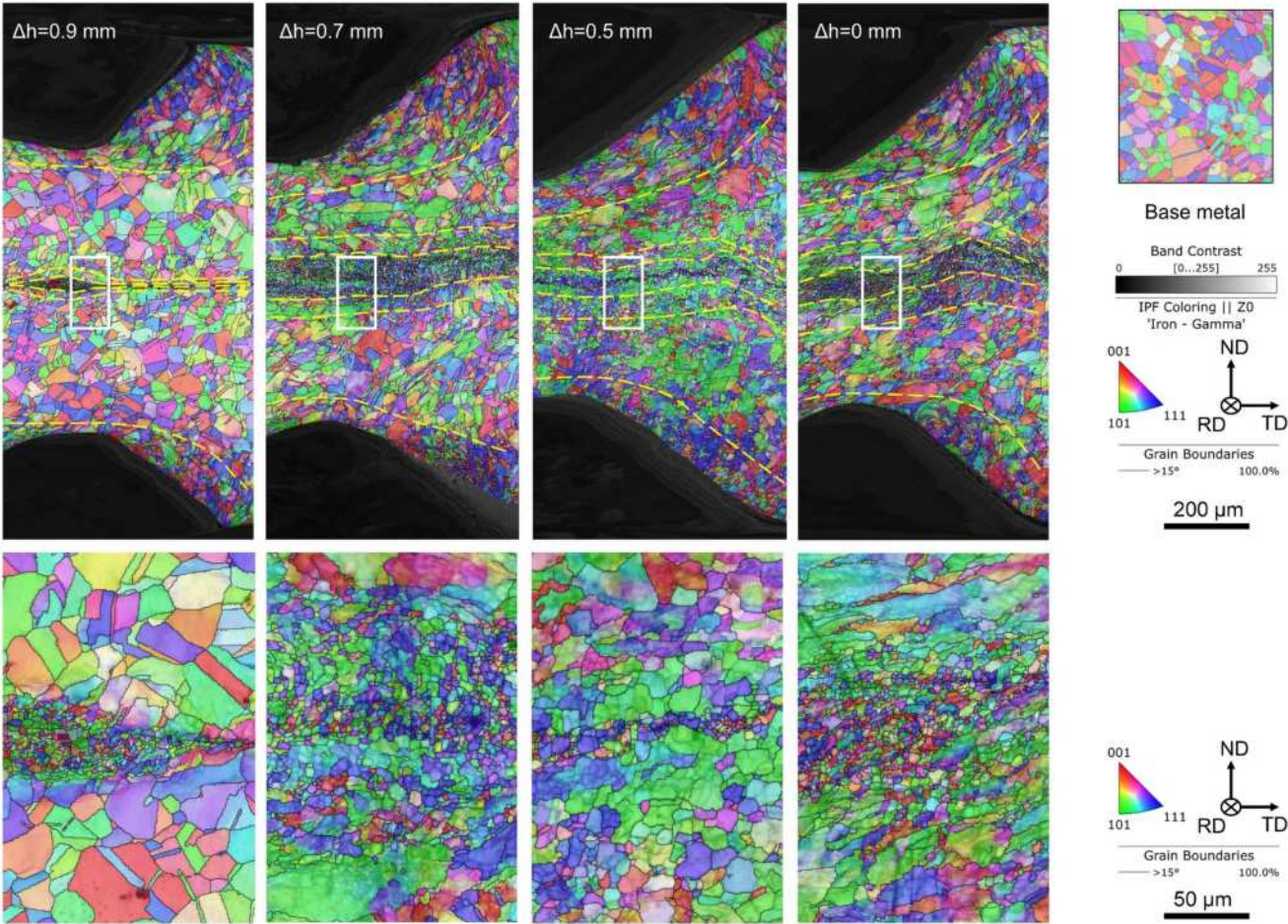
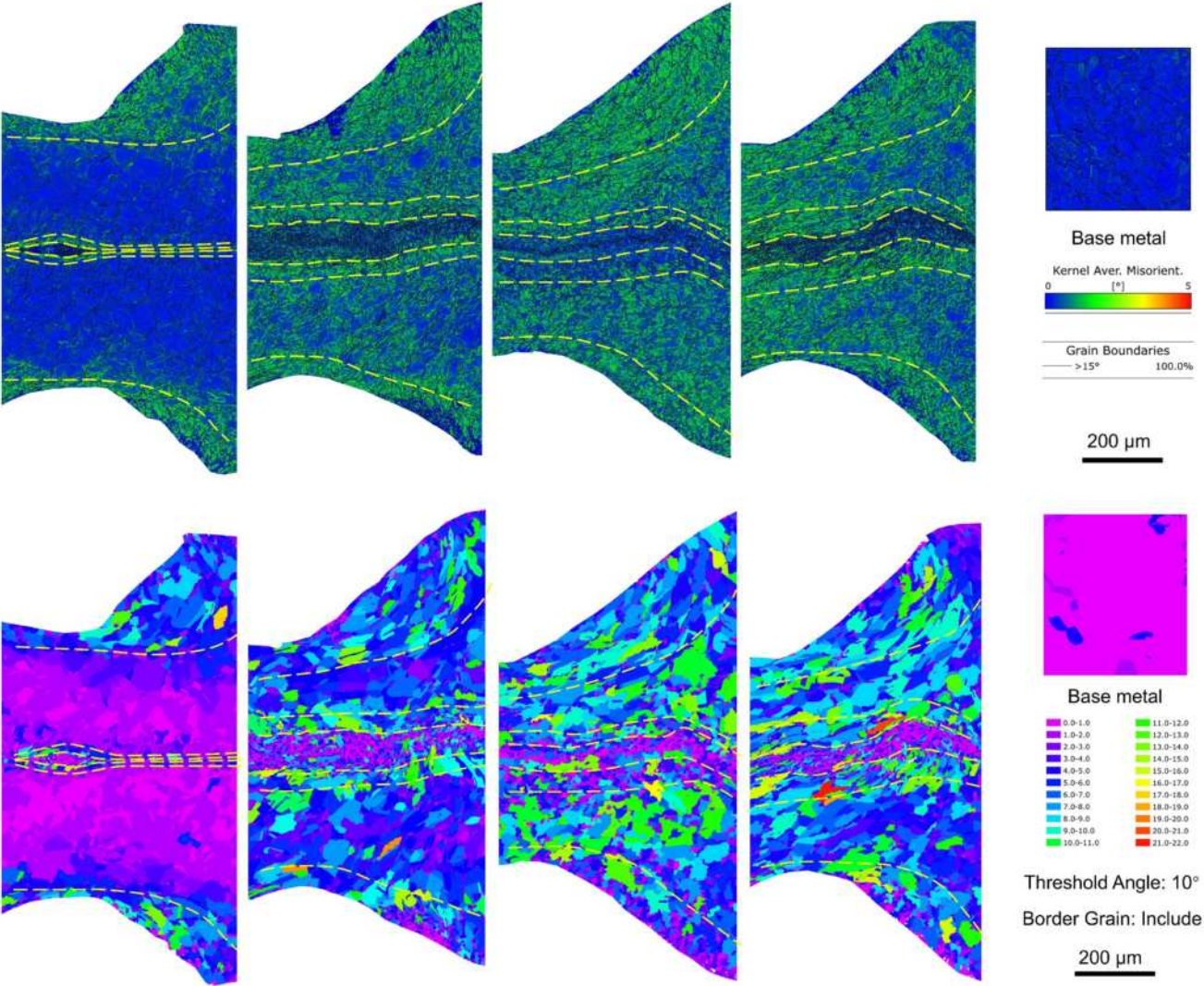
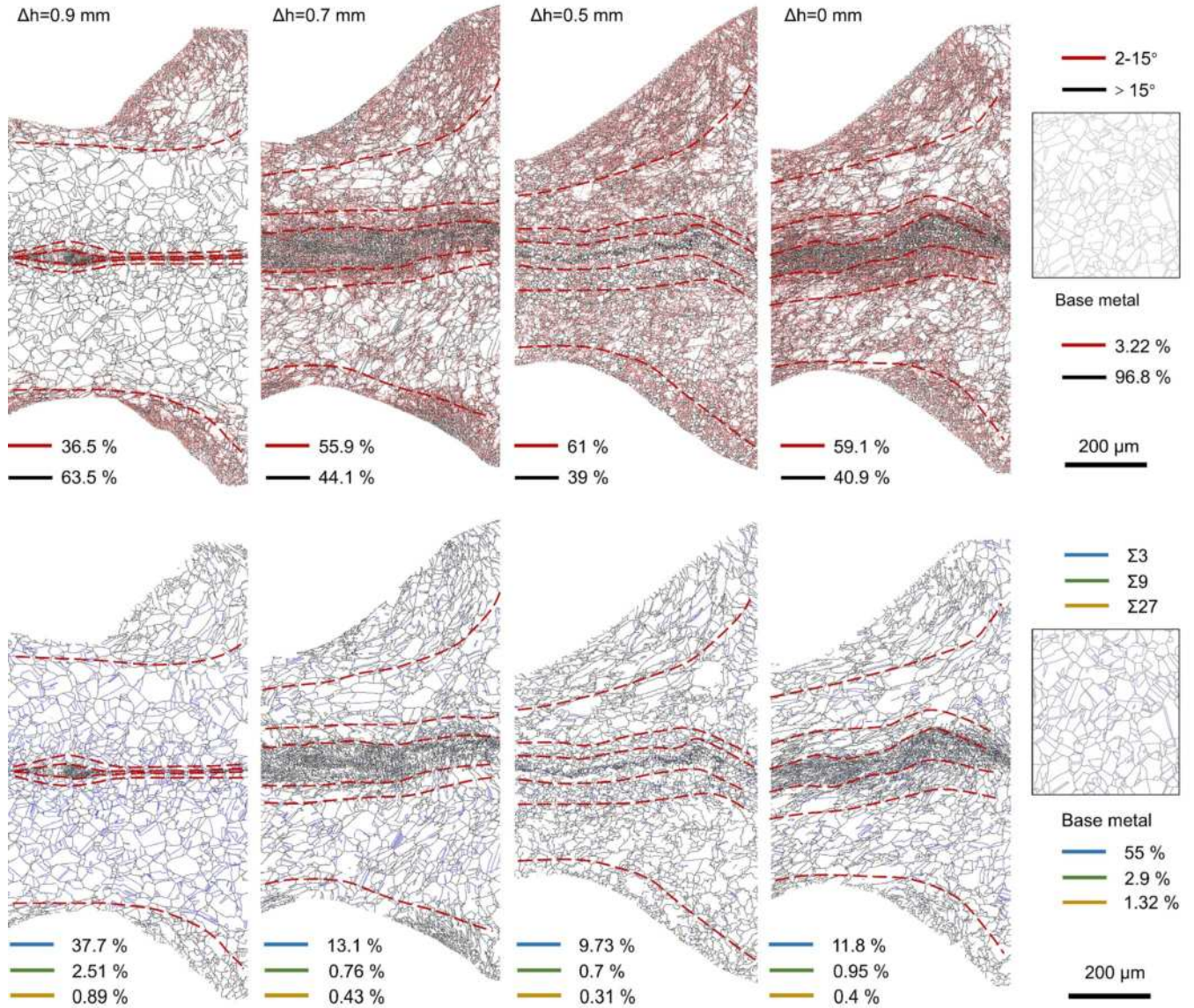


Fig. 13. EBSD characterization of the Invar–Invar weld interface under different preset horn–anvil gaps Δh : IPF maps with band-contrast overlays and HAGBs, together with KAM and GOS maps.



547 **Fig. 13.** (continued).



548

549 **Fig. 14.** Grain-boundary character maps of the interfacial region for the base metal and joints
 550 welded at $\Delta h = 0.9, 0.7, 0.5$ and 0 mm.

551 (1) $\Delta h = 0.9$ mm

552 At $\Delta h = 0.9$ mm, the SAZ contains only narrow refined bands beneath the tool teeth, the
 553 CGTZ remains close to the base metal, and the WS appears only locally at peak-to-peak contact
 554 sites. The seam and adjacent FGTZ are therefore thin and discontinuous, consistent with under-
 555 activation and interface fracture.

556 (2) $\Delta h = 0.7$ mm

At $\Delta h = 0.7$ mm, a continuous fine-grained WS develops along a fully closed interface. The WS is about 90 μm thick with an average grain size of ~ 3.21 μm , the highest LAGB fraction and the lowest $\Sigma 3$ content, giving the most continuous and homogeneous seam in the dataset. Together with the continuous low-KAM band and the predominantly recrystallized GOS response, the $\Delta h = 0.7$ mm boundary state is most consistently interpreted as a mixed recovery-assisted CDRX seam rather than a fully equilibrated recrystallized structure. In the low-stacking-fault-energy Invar system, this state is compatible with high subgrain/dislocation-cell content, progressive twin-network disruption, and improved local deformation compatibility.

(3) $\Delta h = 0.5$ mm

At $\Delta h = 0.5$ mm the interface remains well bonded, but the WS becomes slightly less pronounced. Although the WS thickness is similar to that at 0.7 mm, HAGB fraction increases and LAGB fraction decreases, suggesting a more advanced recrystallisation stage within a more localised seam.

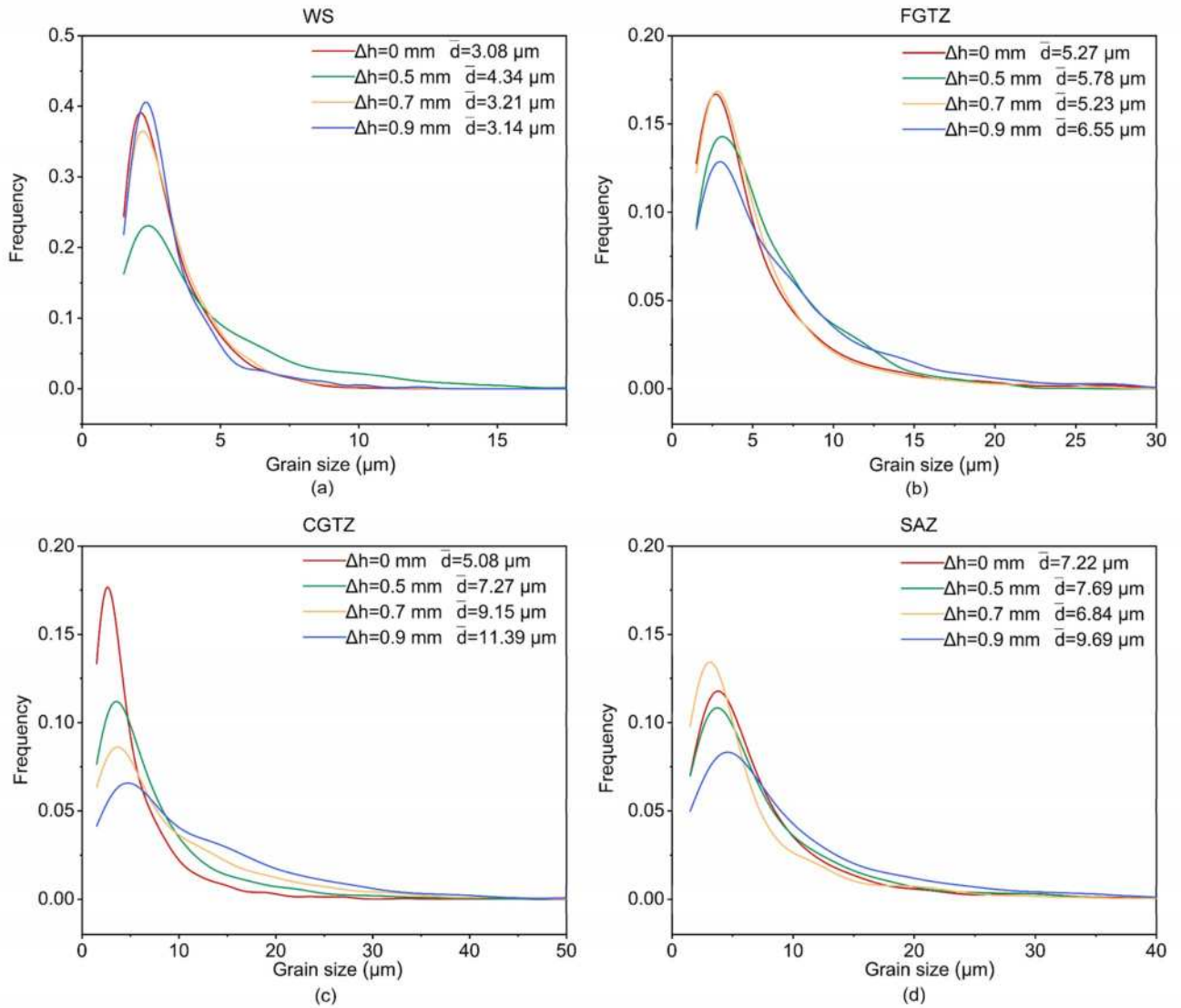
(4) $\Delta h = 0$ mm

At $\Delta h = 0$ mm the WS evolves into a layered streamline-shaped band of very fine equiaxed grains separated by narrow high-KAM ribbons. The seam is therefore not simply finer; it is more stratified and heterogeneous, consistent with repeated late-stage re-shearing rather than monotonic beneficial seam growth.

(5) Overall influence of Δh on interfacial EBSD characteristics

Across all conditions, decreasing Δh from 0.9 to 0.7 mm widens the WS and converts it from local/discontinuous to broad and continuous. Further reduction to 0.5 and 0 mm does not continue this trend; the seam remains fine but becomes more localised or layered, and a larger surrounding volume participates in recovery and rearrangement.

Fig. 15 shows that the WS grain-size distributions are much finer than those of the base metal
 and CGTZ for all welds, with the narrowest and most uniform distribution occurring at $\Delta h = 0.7$
 mm. Because the EBSD step size is $0.6 \mu\text{m}$, these distributions should be read as comparative
 EBSD-resolved statistics rather than as a full census of the finest sub-micrometre features, which
 are additionally supported by the high-magnification SEM/BSE observations in Fig. 12.



585

Fig. 15. Grain-size distributions in the four characteristic zones ((a) WS, (b) FGTZ, (c) CGTZ and
 (d) SAZ) for joints welded at $\Delta h = 0.9, 0.7, 0.5$ and 0 mm.

Table 4. Grain-boundary character distribution in different zones under various Δh conditions.

Δh (mm)	WS HAGB (%)	WS LAG B (%)	WS $\Sigma 3$ (%)	FGTZ HAGB (%)	FGTZ LAG B (%)	FGTZ $\Sigma 3$ (%)	CGTZ HAGB (%)	CGTZ LAGB (%)	CGTZ $\Sigma 3$ (%)
0.9	62.9	37.1	7.86	72.5	27.5	44.1	96.2	3.8	52.2
0.7	35.3	64.7	3.46	35.8	64.2	11.3	48.7	51.3	26.7
0.5	61.0	39.0	7.21	43.5	56.5	10.4	38.6	61.4	10.7
0.0	66.4	33.6	6.79	35.9	64.1	9.58	40.0	60.0	20.5

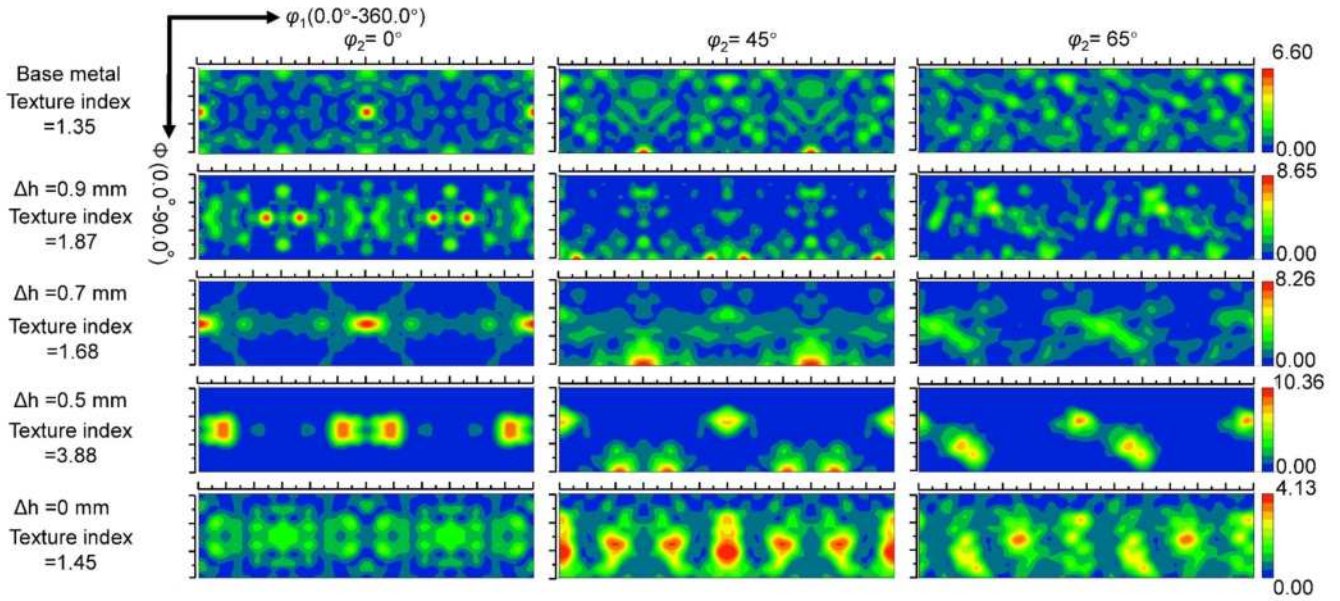
3.5.2 Texture and crystallographic plasticity indicators in the weld seam

Fig. 16 shows ODF sections for the WS and base metal, and Fig. 17 gives the corresponding Schmid-factor m and Taylor-factor M distributions under lap-shear loading. The base metal exhibits a weak recrystallization texture ($J \approx 1.35$, $f_{\max} \approx 6.6$ m.r.d.). At $\Delta h = 0.9$ mm the WS texture strengthens modestly ($J \approx 1.87$, $f_{\max} \approx 8.65$ m.r.d.) without continuous gamma or beta fibres. At $\Delta h = 0.7$ mm the texture remains moderate ($J \approx 1.68$), with strengthened Goss peaks and weak-to-moderate gamma-fibre segments. At $\Delta h = 0.5$ mm the strongest texture is obtained ($J \approx 3.88$, $f_{\max} \approx 10.4$ m.r.d.), with clear shear-texture features dominated by the beta fibre (Brass–S–Copper skeleton) and a secondary gamma contribution. For $\Delta h = 0$ mm the texture weakens again ($J \approx 1.45$, $f_{\max} \approx 4.1$ m.r.d.), consistent with repeated shear–compression–refinement cycles in a layered seam.

For all conditions, Schmid-factor distributions remain concentrated at high m , but the Taylor factor distinguishes the seam states more clearly. The base-metal mean M is about 3.1. After welding, the WS at $\Delta h = 0.7$ mm shows a slightly lower mean M than the base metal, whereas $\Delta h = 0.9$ and 0 mm remain close to the base-metal level. The highest mean M occurs at $\Delta h = 0.5$ mm (about 3.3), indicating the strongest compatibility constraint. These quantitative comparisons show that $\Delta h = 0.7$ mm offers the easiest local accommodation, whereas $\Delta h = 0.5$ mm shifts the seam toward a more constrained state despite remaining within the robust window. Taken together,

the ODF-m-M results distinguish the seam states by their local accommodation signatures: $\Delta h = 0.7$ mm is associated with easier plastic accommodation, $\Delta h = 0.5$ mm with stronger compatibility constraints, and $\Delta h = 0$ mm with a weakened but layered-shear texture state.

610



611

Fig. 16. Selected ϕ_2 sections of the orientation distribution function (ODF) for the base metal and weld seam (WS) under different preset gaps Δh .

613

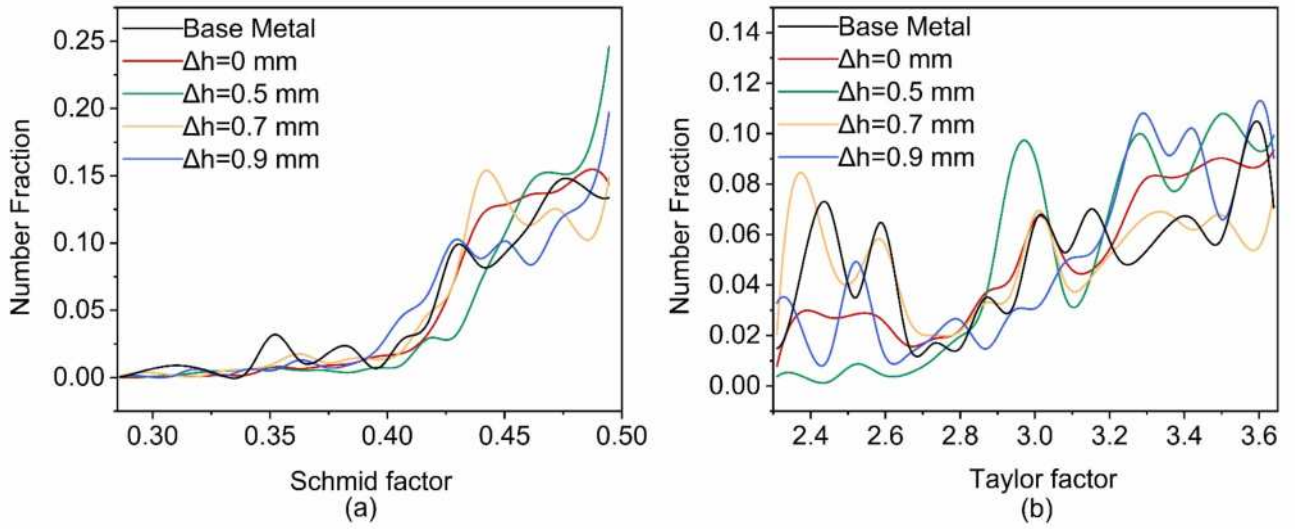


Fig. 17. Distributions of (a) Schmid factor (m) and (b) Taylor factor (M) in the weld seam (WS) for the base metal and joints welded at $\Delta h = 0.9, 0.7, 0.5$ and 0 mm.

3.5.3 Nanoindentation response across the selected WS+FGTZ line

A common 13-point nanoindentation line ($\sim 180 \mu m$ total span, $\sim 15 \mu m$ spacing) was extracted around the seam-centre position for each Δh . For $\Delta h = 0.9$ mm, the nominal WS+FGTZ width is smaller than this common comparison window, so the result should be read as a selected-window comparison rather than a strict seam-width match.

Fig. 18a,b shows a non-monotonic selected-line response. $\Delta h = 0$ mm gives the highest line-mean hardness and reduced modulus together with the smallest h_{max} , whereas $\Delta h = 0.5$ mm gives the lowest H and E_r and the largest h_{max} . $\Delta h = 0.7$ and 0.9 mm have similar mean E_r levels, but $\Delta h = 0.7$ mm exhibits the most compact H/E_r profile. The profile shape further differentiates the local interfacial response. $\Delta h = 0.7$ mm shows the lowest E_r dispersion and a low H dispersion, $\Delta h = 0.5$ mm shows a lower H/E_r response across the selected line, and $\Delta h = 0$ mm combines higher local indentation resistance with the largest spatial fluctuation.

629 Fig. 18c,d shows the same ranking in the full point distributions: $\Delta h = 0.5$ mm is shifted to the
630 low end, $\Delta h = 0$ mm to the high and more scattered end, and $\Delta h = 0.7$ mm forms the most compact
631 intermediate distribution. Although the $\Delta h = 0.9$ mm boxplots partly overlap those at 0.7 mm, the
632 common-window construction and the much narrower nominal WS+FGTZ width indicate that
633 $\Delta h = 0.9$ mm does not provide the same broad and stable local interfacial response.

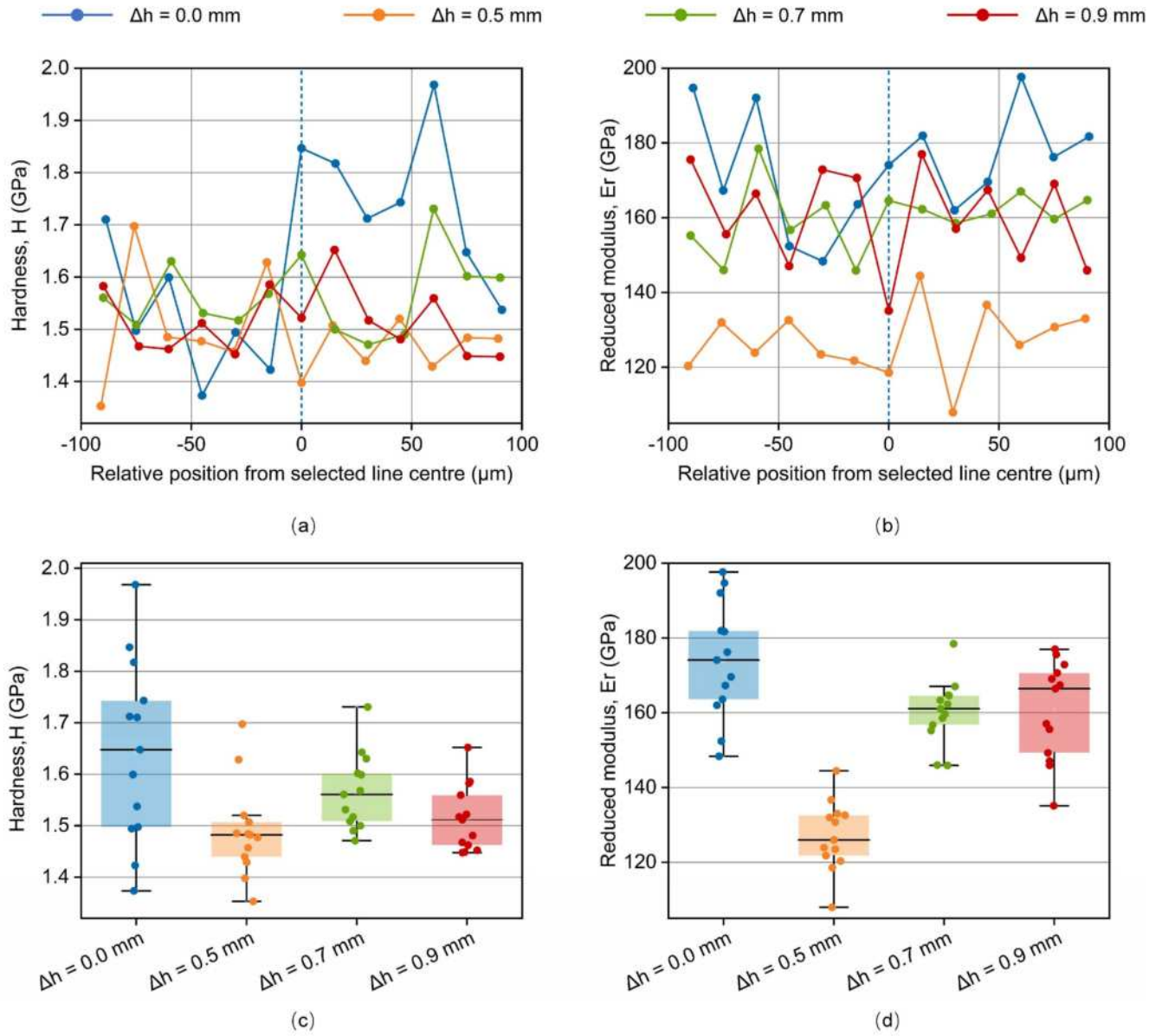


Fig. 18. Nanoindentation response across the selected 13-point interfacial windows for joints welded at different preset gaps. (a) Hardness H and (b) reduced modulus E_r as a function of relative position from the selected line centre. (c) Boxplot of H and (d) boxplot of E_r with all raw points overlaid for the same selected 13-point windows. A common window length ($\sim 180 \mu\text{m}$ total span) was retained for cross-condition comparison.

641 4. Discussion

642 4.1 Δh -controlled loading path and weld-collapse window

643 In the present pneumatic UMW system, the vibration amplitude, nominal weld time, and
 644 preset static normal load are fixed, so Δh is the only deliberately varied machine parameter. Its role
 645 is therefore not to change the imposed oscillatory source itself, but to schedule when the weld
 646 departs from force-dominant indentation and enters a displacement-constrained stage after lower
 647 dead-point engagement. In this sense, Δh should be interpreted as a position-triggered
 648 displacement constraint rather than as a simple geometric stop or a surrogate for final collapse
 649 alone. This framing differs from the conventional time- or energy-defined welding windows used
 650 in ultrasonic spot welding ^[1,6]. The useful Δh window identified below should therefore be
 651 interpreted as condition-specific for the present amplitude, weld time, and machine-compliance
 652 setting rather than as a universal scale transferable without recalibration.

653 Across the present Invar welds, decreasing Δh produces two coupled but non-equivalent
 654 trends: increasing collapse and decreasing late-stage static-force relaxation. These trends define
 655 three distinct process regimes. At $\Delta h = 0.9$ mm, the process remains under-activated: unloading
 656 occurs early, the reconstructed seam is narrow and discontinuous, and interface fracture dominates.
 657 At intermediate gaps (0.7–0.5 mm), sufficient collapse is achieved before late-stage damage
 658 accumulation becomes dominant, producing a robust bounded window. At $\Delta h = 0$ mm, collapse is
 659 no longer the limiting factor; instead, persistent normal constraint and tangential activity promote
 660 re-shearing, thinning, and renewed interface-proximal damage. The following two subsections
 661 separate the time-domain evolution of the static normal load from the space-domain information
 662 contained in the tri-axial force envelopes.

663 4.1.1 Time-domain modulation of $F_{Z,static}(t)$ and weld collapse

664 The evolution of the cycle-averaged static normal force can be rationalised using a reduced
 665 machine-spring/joint description ^[25]:

$$666 F_{Z,static}(t) \approx k_m [\delta_z(t) - u_{joint}(t)], \quad (1)$$

667 Here k_m is the effective stiffness of the machine–fixture system, $\delta_z(t)$ is the commanded horn
 668 stroke, and u_{joint} is the thickness reduction of the lap stack (weld collapse). Introducing Δh defines
 669 an effective lower dead point: once the remaining stack thickness approaches
 670 this dead point, further horn motion is constrained and subsequent
 671 deformation is accommodated mainly by additional collapse together with release of machine
 672 compliance. In the present time-terminated pneumatic platform, Δh therefore does not prescribe an
 673 identical absolute relaxation time in a closed-loop sense; instead, it prescribes the geometric
 674 threshold at which the displacement-constrained stage begins, while the actual clock time follows
 675 from the collapse kinetics of the stack.

$$676 F_{Z,static}(t) \approx F_{Z,0} - k_m \max\{0, u_{joint}(t) - (H_0 - \Delta h)\}, \quad (2)$$

677 In Eq. (2), $F_{Z,0}$ is the initial static normal force and H_0 is the initial total thickness of the lap
 678 stack. The decay of $F_{Z,static}$ should therefore be interpreted primarily as compliance-controlled
 679 unloading under a displacement constraint, rather than as a direct signature of interfacial slip-to-
 680 stick evolution. The relation is used only as a reduced-order process descriptor anchored to the
 681 monitored collapse and force history; it is not intended as a constitutive model of the true local
 682 interfacial stress field.

683 To summarise this loading-path change, two non-dimensional indicators are retained: a
 684 collapse ratio $\phi_c(\Delta h)$ and a force-retention indicator $\eta_F(\Delta h)$. The first captures how much of the
 685 available stack-thickness reduction has actually been consumed, whereas the second captures how

strongly the static normal load is retained during sonics. This separation is important because it distinguishes “how much collapse occurred” from “how long high normal constraint persisted”, which are not the same once Δh becomes small.

$$\phi_c(\Delta h) = u_{horn,z} / H_0, \quad (3)$$

$$\eta_F(\Delta h) = 1 - \Delta F_{Z,static} / F_{Z,0}, \quad (4)$$

For the present welds, the collapse ratio rises from about 0.07 at $\Delta h = 0.9$ mm to ~ 0.22 at 0.7 mm and ~ 0.34 at $\Delta h \leq 0.5$ mm, indicating that the useful collapse window is already largely exhausted by the time the gap is reduced to 0.5 mm. Over the same range, the force-retention indicator increases from about 0.66 to 0.78 and then to ≥ 0.95 , showing that smaller gaps primarily suppress late-stage static-force relaxation and retain near-constant normal pressure for most of the ultrasonic stage.

The contrast between $\Delta h = 0.7$ and 0.5 mm is particularly informative. By $\Delta h = 0.7$ mm, collapse has already approached saturation, yet force retention still increases markedly as Δh is further reduced. This indicates that, once the geometric collapse ceiling is nearly reached, decreasing Δh yields little additional collapse-driven benefit but substantially prolongs the period over which high normal pressure is maintained, thereby increasing termination power and peak temperature. In this sense, the 0.7 and 0.5 mm comparison isolates the co-ordination role of Δh , after collapse is largely sufficient, the main effect of further reducing Δh is to retain constraint for longer rather than to create proportionally greater seam development. This distinction is captured more directly by the normal-force impulse. Two conditions may reach similar final collapse while still spending different fractions of the ultrasonic stage under high normal pressure; the impulse is therefore retained as an integral descriptor of cumulative normal-load dwell during sonics, i.e., how long strong normal constraint is maintained rather than how much final collapse is obtained.

The overall effective normal loading can be characterised by the normal-force impulse

$$W_n(\Delta h) = \int_0^{t_{US}} F_{Z,static}(t; \Delta h) dt. \quad (5)$$

Here $W_n(\Delta h)$ is the normal-force impulse and t_{US} is the ultrasonic vibration duration. The impulse therefore serves as an integral descriptor of cumulative normal-load dwell during sonics, i.e. how long strong normal constraint is maintained rather than how much final collapse is obtained. Accordingly, $\Delta h = 0.9$ mm gives a condition with small collapse, early unloading, and an undersized discontinuous seam. Reducing Δh to 0.7 mm produces sufficient collapse to disrupt surface films and build a continuous WS/FGTZ network while still preserving some late static-force relaxation. Further reduction toward 0 mm yields little additional collapse but sustains high normal force and higher thermal input, which is consistent with thinning, renewed interface-related failure, and over-processing. In this sense, Δh does not simply set a collapse ceiling; it also schedules how long the latter part of the weld remains in a high-pressure, damage-prone state.

4.1.2 Spatial modulation of tri-axial high-frequency forces

Because the force sensor is installed beneath the anvil, the measured tri-axial signals are transmitted machine-side reactions of the horn–sheet–anvil–fixture system rather than one-to-one measures of the true local interfacial stress state. Recent anvil-integrated force-sensing work similarly treats these signals as valuable comparative monitoring signatures, while still emphasizing the effects of transfer behaviour, structural dynamics and damping on exact force quantification [26,27]. By contrast, direct relative-motion or bond-formation kinematics require dedicated optical methods such as PDV or LDV [11,12]. Accordingly, quantities such as $\tilde{F}_{Y,min}$ and

730 $\tilde{F}_{X,max}$ are interpreted here as bounded machine-side response descriptors under a fixed machine
 731 architecture, not as direct local stress measurements.

732 With vibration amplitude and frequency fixed, the imposed oscillatory displacement can be
 733 treated as constant to first order. The late-stage force response therefore contains two coupled
 734 contributions. One is compliance-controlled: once Δh constrains further horn sinking, release of
 735 static normal load reduces the normal component of the transmitted dynamic force. The other is
 736 interface-related: the tangential response relaxes if the load-bearing interface has evolved toward a
 737 more stable late-stage state, but remains high if late-stage re-shearing and over-constrained contact
 738 persist. Δh affects both contributions simultaneously by changing when the displacement-
 739 constrained stage begins and how much normal and tangential constraint survive after that point.

$$740 \quad \tilde{F}_Y(t; \Delta h) \approx k_t(\Delta h, t) u_{vib}, \tilde{F}_Z(t; \Delta h) \approx k_n(\Delta h, t) u_{vib}, \quad (6)$$

741 where u_{vib} is the vibration displacement amplitude and k_t , k_n are effective tangential and
 742 normal contact stiffnesses. In this sense, $\tilde{F}_{Y,min}$ should not be read in isolation. For the gap-
 743 constrained conditions ($\Delta h = 0.9\text{--}0.5$ mm), $\tilde{F}_{Y,min}$ remains in a similarly low band, but the physical
 744 meaning is not the same. At $\Delta h = 0.9$ mm, the low late-stage tangential response is associated
 745 mainly with premature unloading and insufficient interface development; this is consistent with the
 746 smallest collapse, the lowest termination power, and the dominance of interface fracture. At $\Delta h =$
 747 $0.7\text{--}0.5$ mm, by contrast, low $\tilde{F}_{Y,min}$ is observed after sufficient interfacial activation has already
 748 occurred, so it is more consistent with genuine late-stage shear relaxation under an established
 749 seam. The comparison between $\Delta h = 0.5$ and 0 mm is even more informative: both conditions show
 750 little static-force relaxation, yet only $\Delta h = 0$ mm exhibits an abnormally high late-stage tangential
 751 response. Persistent high $\tilde{F}_{Y,min}$ therefore signals continued re-shearing and damage accumulation
 752 rather than retained normal load alone.

The transverse component $\tilde{F}_x$ provides an additional discriminator. Although the macroscopic vibration is along Y, substantial $\tilde{F}_x$ values are expected because tooth-scale indentation, lateral extrusion and folding are inherently three-dimensional. Intermediate gaps show relatively low $\tilde{F}_{x,max}$, consistent with partial relaxation of off-axis constraint once a continuous seam has formed. Zero gap, however, gives the highest $\tilde{F}_{x,max}$, consistent with persistent lateral confinement and repeated folding/re-shearing in the seam region.

Taken together, the force envelopes show that Δh coordinates not a single load level but the coexistence window of collapse, static-force retention and late-stage tangential activity. That is why the preset horn–anvil gap can separate bonding-dominant and damage-dominant stages even when amplitude, preset load and nominal weld time are held constant.

4.2 Coupled effect of preset gap Δh on microstructural evolution in the weld seam and joint fracture behaviour

Although peak load reaches a plateau at intermediate Δh , fracture displacement and failure mode still vary markedly. This divergence reflects the coupled dependence of weld-seam microstructure and damage evolution on the full thermal–mechanical loading path: Δh reshapes the static normal-force history $F_{Z,static}(t)$, the dynamic force-envelope histories in Y and Z, and the temperature history, thereby controlling the competition between dynamic recovery (DRV), dynamic recrystallisation (DRX/CDRX), grain growth and damage accumulation in the interfacial region.

Invar 36 is an fcc austenitic alloy that, in the annealed condition, contains a high density of $\Sigma 3$ annealing twins (Fig. 14). Under severe shear, plastic accommodation can proceed by dislocation slip combined with detwinning and subgrain formation; with increasing temperature and accumulated strain, recovery-assisted continuous DRX and, locally, discontinuous DRX can

776 occur, producing the ultrafine equiaxed grains observed in the WS. Therefore, the evolution of
 777 boundary character (HAGB/LAGB/ $\Sigma 3$ fractions), KAM/GOS fields and texture components
 778 provides a mechanistic bridge between the monitored loading path and macroscopic joint
 779 performance.

780 4.2.1 Thermal–mechanical window based on a relative-Z indicator: Δh -controlled DRV/DRX
 781 competition and WS length-scale evolution

782 UMW rapidly raises local temperature through coupled frictional and plastic dissipation ^[1,2].
 783 Here the Zener–Hollomon framework is used only comparatively to rank seam states under the
 784 present machine architecture.

$$785 \quad Z = \dot{\epsilon} \exp\left(\frac{Q}{RT}\right) \quad (7)$$

786 where $\dot{\epsilon}$ is the equivalent strain rate, Q the activation energy, R the gas constant, and T the
 787 absolute temperature. Because the true interface temperature and absolute interfacial strain rate
 788 cannot be measured directly, the objective is not to calculate an absolute Z value, but to retain a
 789 weld-seam-based indicator that preserves cross-condition trends ^[28].

790 For a shear-dominated seam, a characteristic strain-rate scale may be written as

$$791 \quad \dot{\gamma}_{WS} \sim \omega A_{eff} / \delta_{WS}, \omega = 2\pi f \quad (8)$$

792 where f is the vibration frequency, A_{eff} an effective sliding amplitude, and δ_{WS} an effective
 793 shear-layer thickness. Because A_{eff} and δ_{WS} are not available in situ, the monitored tangential
 794 envelope F_Y is used as a practical proxy for shear-related interface activity under fixed amplitude
 795 ^[3,25].

796 To compare different Δh conditions, the rise–peak–decay evolution of F_Y within the effective
 797 ultrasonic window is retained by using a two-stage average tangential envelope:

$$\tilde{F}_{Y,avg} \approx (\tilde{F}_{Y,I} t_1 + \tilde{F}_{Y,II} t_2) / (t_1 + t_2), t_2 = t_{US} - t_1 \quad (9)$$

$$\tilde{F}_{Y,I} = (\tilde{F}_{Y,0} + \tilde{F}_{Y,max}) / 2, \tilde{F}_{Y,II} = (\tilde{F}_{Y,max} + \tilde{F}_{Y,min}) / 2 \quad (10)$$

where $\tilde{F}_{Y,0}$ is the initial envelope amplitude at the start of the effective window. Combined with the measured seam width, this yields a relative shear-intensity indicator $\dot{\gamma}_{ws}^i$ for the weld seam.

$$\dot{\gamma}_{ws}^i = \tilde{F}_{Y,avg} / w_{ws} \quad (11)$$

This quantity should be read only as a comparative measure of shear activity per unit seam width; it is not an absolute strain rate.

Table 5. Integrated metrics linking average tangential loading, weld-seam width, refined interfacial-region width, relative shear intensity, and weld-seam grain size under different Δh .

$\Delta h(\text{mm})$	$\tilde{F}_{Y,avg}(\text{N})$	$w_{ws}(\mu m)$	$\dot{\gamma}_{ws}^i(\text{N}/\mu m)$	$w_{ws} + w_{FGTZ}(\mu m)$	$d_{ws}(\mu m)$
0.9	127	28	4.54	66	3.14
0.7	163	91	1.79	180	3.21
0.5	160	75	2.13	172	4.34
0.0	237	68	3.49	179	3.08

Table 5 highlights the end-member contrast: $\Delta h = 0.9$ mm gives the lowest shear-intensity indicator and an extremely narrow WS, whereas $\Delta h = 0$ mm combines the highest indicator with a narrower layered seam.

This distinction becomes clearer when Table 5 is read together with the microstructural and nanoindentation results. At $\Delta h = 0.9$ mm, the low shear-intensity indicator and $\sim 28 \mu m$ WS imply that only a very small reconstructed seam volume is created. The discontinuous WS, dominant interface fracture, and low local E_r near the seam centre are therefore more consistent with an undersized seam than with a stable continuous one. At $\Delta h = 0.7$ mm, w_{ws} expands to $\sim 91 \mu m$ and the relative shear-intensity indicator reaches its lowest value among the gap-constrained cases. Together with the highest WS LAGB fraction, the lowest $\Sigma 3$ fraction, the narrowest WS grain-size

817 distribution, and the most compact selected-line H/E_r response, this condition is most consistent
 818 with the most favourable continuous fine-grained WS morphology in the present dataset. However,
 819 the shear-intensity indicator alone still captures only the mechanical side of the seam state. To
 820 compare whether a given WS condition is tilted more toward recovery/recrystallisation or toward
 821 localisation under the present material system, a temperature-corrected relative indicator is
 822 introduced by combining the seam-scale shear activity with a characteristic seam temperature.

823 On this basis, a relative Zener–Hollomon indicator for the WS is defined as

$$824 \quad Z_{WS}^{\dot{\epsilon}} = \dot{\gamma}_{WS}^{\dot{\epsilon}} \exp\left(\frac{Q}{RT_{WS}}\right) \quad (12)$$

825 where T_{WS} is a characteristic seam temperature. The weld-centre peak temperature is used as a
 826 consistent approximation of T_{WS} for cross-condition comparison ^[2,9]. To avoid assuming a
 827 specific activation energy, a threshold activation energy that equalises two adjacent relative
 828 indicators can be written as

$$829 \quad Q_{th} = R \ln\left(\dot{\gamma}_{WS,1}^{\dot{\epsilon}} / \dot{\gamma}_{WS,2}^{\dot{\epsilon}}\right) / \left(1/T_2 - 1/T_1\right) \quad (13)$$

830 For the two adjacent comparisons considered here, this yields

$$831 \quad Q_{th,0.7 \rightarrow 0.5} \approx 7 \text{ kJ mol}^{-1}, Q_{th,0.5 \rightarrow 0} \approx 74 \text{ kJ mol}^{-1} \quad (14)$$

832 The much smaller threshold for $0.7 \rightarrow 0.5$ mm than for $0.5 \rightarrow 0$ mm shows that the first step is
 833 governed mainly by temperature rise within a still coherent seam, whereas the second reflects a
 834 stronger contribution from strain-rate concentration and late-stage shear localisation. Such a
 835 localization-dominated response is consistent with previous studies showing that ultrasonic
 836 vibration can intensify local plastic deformation and alter refinement behavior during
 837 compression^[29].

838 A second way to describe seam evolution is through the relation between WS width and the
 839 combined WS + FGTZ width. For $\Delta h = 0.7, 0.5$ and 0 mm, WS + FGTZ remains nearly constant
 840 while the WS itself shrinks from $\sim 91 \mu\text{m}$ to a much narrower layered band. Once a continuous
 841 seam has formed, further input therefore promotes seam subdivision, local re-shearing, and
 842 heterogeneity rather than simple seam widening.

843 The monitoring signatures are consistent with this picture. For the gap-constrained conditions
 844 ($\Delta h \geq 0.5$ mm), $\tilde{F}_{y,min}$ relaxes to a low level, but at $\Delta h = 0$ mm it remains high even after collapse
 845 saturation, indicating that added input is dissipated through continued indentation and repeated
 846 local shearing rather than through beneficial seam development.

847 Overall, Δh defines three regime classes: an under-activated state at 0.9 mm, a robust
 848 intermediate-gap state at 0.7 – 0.5 mm, and an over-processed layered state at 0 mm. Within the
 849 intermediate class, $\Delta h = 0.7$ mm gives the broader and more continuous seam, whereas $\Delta h = 0.5$
 850 mm remains robust but is narrower and more localized. The useful window is therefore governed
 851 by how Δh balances seam formation against late-stage reprocessing rather than by collapse alone.

852 4.2.2 Divergence of fracture mode and ductility: combined effect of WS length-scale threshold and
 853 texture/local constraint

854 Lap-shear results place the mechanical optimum in an intermediate-gap window ($\Delta h = 0.5$ –
 855 0.7 mm). These conditions are best read as neighbouring states within a robust high-performance
 856 regime rather than sharply separated optima.

857 At the first level, suppression of interface fracture depends mainly on whether a continuous,
 858 sufficiently wide reconstructed interfacial region forms. At $\Delta h = 0.9$ mm, low collapse, weak
 859 tangential activation and early unloading coincide with a discontinuous WS and limited FGTZ. By

contrast, at $\Delta h = 0.7$ mm the WS and FGTZ become continuous and sufficiently wide to shift failure away from the interface.

Zero gap shows why seam continuity alone is insufficient. Once a continuous seam has formed, additional ultrasonic input is increasingly dissipated through repeated local shearing, continued indentation and damage accumulation. The multilayer ultrafine bands at $\Delta h = 0$ mm are therefore interpreted as local ultrafine recrystallisation embedded in damage-associated localisation bands.

Once interface fracture is largely suppressed, the next level is controlled by strain localisation and damage tolerance in the TTT zone. At $\Delta h = 0.7$ mm, the seam shows a broad continuous WS, the highest WS share within the refined region, and a relatively low mean Taylor factor, while the selected-line nanoindentation profile remains the most compact among the tested conditions. In this low-stacking-fault-energy Invar system, the LAGB-rich and $\Sigma 3$ -depleted boundary state is most consistently read as a recovery-assisted CDRX seam in which boundary subdivision and partial twin consumption improve compatibility only while late-stage re-shearing remains controlled. By contrast, $\Delta h = 0.5$ mm remains within the same high-performance window but shows a narrower, more localised seam together with a lower selected-line H/E_r response, whereas $\Delta h = 0$ mm shows higher local indentation resistance but much stronger spatial heterogeneity along the selected line. Mechanistically, the WS–FGTZ–base-metal hierarchy should be viewed as a strain-transfer architecture rather than as a simple bonded-area effect. The selected-line nanoindentation profile therefore serves only as supporting evidence for this hierarchy: at $\Delta h = 0.7$ mm the broader and more continuous seam promotes more gradual strain transfer into the adjacent thinned zone, whereas at $\Delta h = 0.5$ mm the seam remains strong but more

882 localized, and at $\Delta h = 0$ mm the layered heterogeneous seam promotes damage hot spots instead of
 883 deformation-compatible load transfer.

884 To relate grain refinement and orientation constraint on a common scale, Hall–Petch and
 885 Taylor frameworks may be combined qualitatively as ^[30]:

$$886 \quad \sigma_{flow} \propto M \left(\sigma_0 + k d_{ws}^{-1/2} \right) \quad (15)$$

887 where d_{ws} is the weld-seam grain size, M is the Taylor factor, and σ_{flow} and k are material
 888 constants. The relation is used here only qualitatively to compare how grain refinement and
 889 orientation constraint may co-vary; it is not used to calculate an absolute flow stress. Any Hall–
 890 Petch/Taylor strengthening estimate should therefore be read only qualitatively. $\Delta h = 0$ mm shows
 891 the highest selected-line H/E_r response without improving peak load or fracture displacement,
 892 whereas $\Delta h = 0.5$ mm remains within the high-performance window despite the lowest selected-
 893 line H/E_r .

894 The role of Δh is therefore not to maximise collapse or hardening, but to place the seam above
 895 the bonding threshold while avoiding late-stage re-shearing. Large gaps under-activate the
 896 interface; intermediate gaps create a continuous WS/FGTZ network with manageable
 897 compatibility gradients; and zero gap pushes the weld into over-processing.

898 Pairwise statistical comparison further indicates that the mean peak load and fracture
 899 displacement at $\Delta h = 0.5$ and 0.7 mm are not meaningfully separated in the present dataset; the
 900 practical preference for $\Delta h = 0.7$ mm therefore arises from lower process demand and lower
 901 observed interface-related failure risk, rather than from a distinctly higher mean strength or
 902 ductility. For the present amplitude and weld time, $\Delta h = 0.5$ – 0.7 mm is the recommended window,
 903 with $\Delta h = 0.7$ mm the practical preference because it eliminates observed interface fracture and

combines a broad continuous seam with the most compact local-response distribution at lower power demand than $\Delta h = 0.5$ mm.

4.2.3 Monitoring-driven fracture-mode screening using compact in-process indices

These results imply that in-process screening must detect both under-bonding and abnormal late-stage re-shearing. Because P_{term} overlaps across fracture modes, it was combined with collapse and force-envelope features to build a conservative rule tied to the loading-path interpretation established above. The screening framework uses only signals already measured in the present study, namely P_{term} , final collapse $u_{horn,z}$, static normal-force drop $\Delta F_{Z,static}$, and the early/late tangential envelopes $\tilde{F}_{Y,max}$ and $\tilde{F}_{Y,min}$ (monitored subset $n = 40$; 10 per Δh). Because power alone is ambiguous, the signals were organised into two compact indices representing two complementary process dimensions: I_{bond} , which represents bonding/collapse tendency, and μ_{late} , which represents late-stage shear-to-normal coupling. Their definitions are given in Eqs. (16)–(22).

Let $F_{Z,0}$ be the initial static normal force (constant at 1380 N in our system), and let $F_{Z,end}$ be the static normal force at termination, then:

$$F_{Z,end} = F_{Z,0} - \Delta F_{Z,static} \quad (16)$$

Define the normal-force retention ratio:

$$\eta_F = F_{Z,end} / F_{Z,0} \quad (17)$$

Define the late-to-early dynamic shear retention ratio:

$$\eta_Y = \tilde{F}_{Y,min} / \tilde{F}_{Y,max} \quad (18)$$

Define the relative shear-versus-normal evolution index:

$$\Psi = \eta_Y / \eta_F \quad (19)$$

Ψ compares shear retention with normal-force retention and separates balanced evolution from shear-dominant late-stage behaviour.

927 Define a collapse normalisation term:

$$928 \quad \phi_u = u_{horn,z} / u_{ref} \quad (20)$$

929 Here $u_{ref} = 0.36$ mm, the maximum observed final collapse within the monitored subset.

930 We then define a bonding tendency index:

$$931 \quad I_{bond} = \phi_u (1 - \Psi) \quad (21)$$

932 and a late-stage tangential-to-normal coupling index:

$$933 \quad \mu_{late} = \tilde{F}_{Y,min} / F_{Z,end} \quad (22)$$

934 In this formulation, I_{bond} increases with larger collapse and more consistent normal/shear
 935 evolution, whereas μ_{late} is sensitive to abnormal retained late-stage tangential activity. Together
 936 with P_{term} , the two indices add the state information that power alone does not provide.

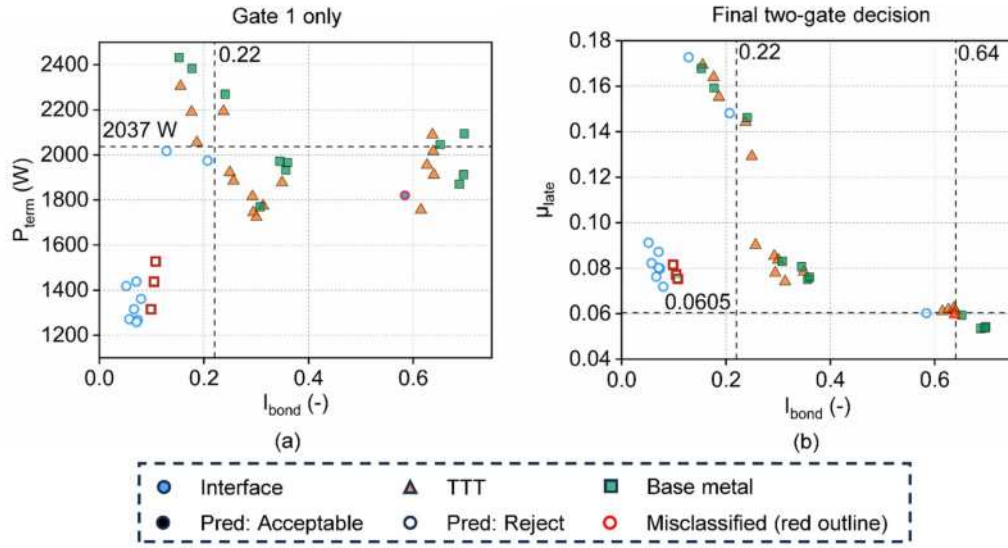
937 Interface-related failure was treated as the safety-critical Reject class, whereas TTT and base-
 938 metal fractures were merged as Acceptable because both require a sufficiently strong interface.
 939 The resulting classifier is intentionally conservative:

940 Gate 1 (under-bonding): Reject if $I_{bond} \leq 0.22$ AND $P_{term} \leq 2037$ W, shown in Fig. 19a.

941 Gate 2 (late-stage anomaly): Reject if $\mu_{late} \leq 0.0605$ AND $I_{bond} \leq 0.64$, shown in Fig. 19b.

942 Accept otherwise (predicted non-interface fracture: TTT + base metal).

943 In the monitored subset ($n = 40$), the two-gate rule gives 90.0% accuracy (36/40), with zero
 944 false acceptance of interface fractures and four conservative false rejections. Gate 2 is retained
 945 because Gate 1 alone would falsely accept 1/10 interface fractures.



946

947 **Fig. 19.** Monitoring-driven two-gate binary screening of interface-related failure risk using in-
 948 process indices (representative monitored subset: $n = 40$; 10 specimens per Δh). (a) Gate 1
 949 visualization in (I_{bond}, P_{term}) , targeting under-bonding. (b) Final binary accept/reject visualization
 950 in (I_{bond}, μ_{late}) , after Gate 1 has removed the low-bonding cluster.

951 Direct transfer of these thresholds to an altered welding-time condition at $\Delta h = 0.5$ mm is
 952 not valid, because the preliminary weld-time perturbation check changed the mapping between
 953 the monitored features and the observed fracture outcome. The present rule should therefore be
 954 read as a condition-specific screening architecture rather than a universal threshold set, and
 955 recalibration is required when the welding recipe changes.

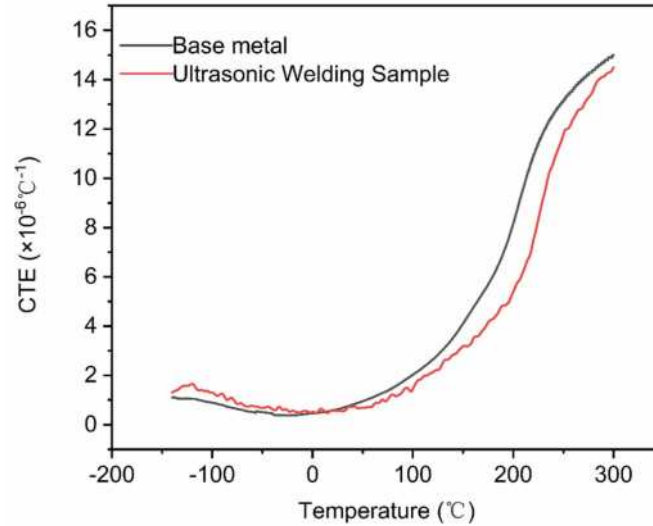
956 4.3 Short-term Curie-temperature excursion and implications for dimensional stability

957 At $\Delta h = 0$ mm, the weld-centre peak temperature reaches ~ 357 °C, above the reported Curie-
 958 temperature interval of Invar 36^[17]. Because the Invar effect is magnetically derived^[16,31], such a
 959 brief local excursion is expected to suppress the low-CTE state transiently during welding. In the
 960 present sub-second thermal cycle, the more immediate concern is therefore not bulk phase
 961 instability but transient thermal-expansion mismatch and residual-stress evolution between the hot
 962 seam and the colder constrained surroundings. The zero-gap seam state points in the same

963 direction: it combines the highest peak temperature with persistent late-stage re-shearing, layered
964 heterogeneity, and reduced fracture displacement. This combination is more consistent with
965 transient thermal-expansion mismatch and residual-stress evolution than with an obvious
966 permanent loss of the macroscopic low-CTE response.

967 Dilatometry was therefore performed on a worst-case $\Delta h = 0$ mm coupon welded under the
968 same nominal parameters. Multiple spot welds were arranged in parallel to form a continuous strip,
969 from which a 10 mm $\times$ 3 mm gauge region was cut. The CTE was measured under flowing N₂ from
970 -140 to 300 °C at 5 °C min⁻¹, with an as-received base-metal coupon tested under the same
971 program. Fig. 20 shows that the overall CTE-temperature responses remain close and that the low-
972 CTE regime around ambient temperature is retained.

973 This result remains macroscopic in scope. The measurement is $n = 1$ and cannot resolve subtle
974 changes within the ultrafine interfacial layer. It therefore does not exclude all Invar-related risk, but
975 it indicates that the short zero-gap excursion did not cause an obvious permanent loss of the
976 macroscopic low-CTE response. From a process-design perspective, $\Delta h = 0.7$ mm remains the
977 conservative choice.



978

979 **Fig. 20.** Measured CTE–temperature curves of the Invar 36 base metal and the $\Delta h = 0$ mm
 980 ultrasonic-welded coupon ($n = 1$, dilatometry, -140 to 300 °C, 5 °C min^{-1} , N_2).

981 4.4 Engineering considerations for industrial deployment of preset-gap-constrained

982 UMW

983 Although the present study isolates the physical role of Δh under laboratory conditions,
 984 industrial deployment in a high-cycle environment would require the manual procedure to be recast
 985 as a calibrated positioning strategy rather than a nominal nut setting. In production, Δh should be
 986 referenced to a verified lower dead point defined at a prescribed static load and then applied as a
 987 recipe-controlled offset. The engineering target is therefore not the nominal gap itself, but the
 988 repeatable onset of displacement-constrained loading under a given machine configuration. In
 989 practice, the current double-nut stop would be replaced by an indexed electromechanical hard stop,
 990 a low-backlash motorized z-offset, or a servo-assisted zeroing routine integrated into the welder.

991 Tool wear introduces the main obstacle to maintaining such equivalence. As the knurled horn
 992 teeth wear, tooth height, local contact stiffness, and frictional engagement all evolve, so a
 993 nominally constant Δh will no longer produce the same lower-dead-point engagement or the same

late-stage loading path. Practical deployment would therefore require scheduled inspection, regrounding or replacement, and periodic re-baselining using reference gauges, verification coupons, or an equivalent calibration routine. Because the monitoring signatures drift with tooling condition, this maintenance should be coupled with monitoring-assisted drift detection. Collapse-related I_{bond} , the late-stage anomaly index μ_{late} , together with power and force-envelope descriptors, could be used as sentinels to trigger recalibration before the process drifts into underbonding or re-shearing-dominated states.

From an automation perspective, the present double-nut arrangement should therefore be regarded as a laboratory implementation of the gap-control principle rather than production hardware. On an automated line, the same principle is more likely to be implemented through a motorized hard stop, a servo-defined lower dead point, or a pneumatic-servo hybrid module with stored recipes, interlocks, and traceable changeover records. Such architectures increase hardware and integration cost relative to a manual stop, but they offer better repeatability, faster recipe changeover, and clearer process traceability. Their long-term economics, reliability, and maintenance burden were not evaluated here, so they are best viewed as engineering implications and future-work directions rather than validated industrial outcomes.

1010 **Conclusions**

In pneumatic ultrasonic metal welding of high-strength, low-ductility sheets, a preset horn–anvil gap acts as a position-triggered displacement constraint. Once the lower dead point is engaged, the process switches from force-dominant indentation to displacement-constrained loading with static-force relaxation while ultrasonic vibration continues. By scheduling the onset of load relaxation relative to weld collapse, the preset gap reshapes normal-force retention, late-stage tangential response, and thermal input, thereby separating bond growth from damage

1017 accumulation. The preset gap should therefore be treated as a physically meaningful loading-path
1018 variable rather than a simple machine setting.

1019 Under the present condition set, three regimes emerge: large-gap under-loading, an
1020 intermediate robust window, and zero-gap over-processing with persistent re-shearing and damage
1021 accumulation. $\Delta h = 0.5\text{--}0.7$ mm defines the robust window, with $\Delta h = 0.7$ mm the practical
1022 preference because it suppresses interface fracture at lower process demand than $\Delta h = 0.5$ mm.
1023 This window should be regarded as condition-specific rather than universal, since its position is
1024 expected to shift with amplitude, weld time, tool state, and machine compliance.

1025 Joint integrity is governed by seam architecture and local compatibility rather than bonded
1026 area alone. Near $\Delta h = 0.7$ mm, a broad continuous weld-seam plus fine-grained transition-zone
1027 network forms together with a recovery-assisted continuous dynamic recrystallization seam and
1028 the most uniform local mechanical response, supporting high displacement capacity and stable
1029 non-interface failure. At $\Delta h = 0.5$ mm, the joint remains within the same high-performance window
1030 but the seam becomes narrower and more constrained, whereas at $\Delta h = 0$ mm persistent late-stage
1031 re-shearing produces seam stratification, spatial heterogeneity, and interface-proximate damage
1032 despite high local indentation resistance.

1033 For Invar-specific functionality, the zero-gap condition raises the weld-centre temperature
1034 above the reported Curie range and is therefore undesirable from a thermal-exposure standpoint.
1035 However, the present post-weld dilatometry did not show a discernible permanent loss of the
1036 macroscopic low-CTE response after this short excursion. A practical in-process screening
1037 framework is achieved by combining termination power with collapse and force-retention metrics
1038 rather than relying on power alone. More broadly, preset-gap scheduling provides a practical basis
1039 for loading-path design and in-process screening in pneumatic ultrasonic metal welding of high-

1040 strength, low-ductility sheet alloys. Future work should extend this concept to closed-loop
 1041 implementation and to dimensionless displacement-constraint metrics that transfer more readily
 1042 across amplitudes, weld times, thicknesses, and tooling conditions.

1043 **Declaration of generative AI and AI-assisted technologies in the writing process**

1044 During the preparation of this work the authors used ChatGPT (OpenAI) to assist with
 1045 language polishing and improving the clarity of some sentences. ChatGPT was also used for
 1046 information gathering and analysis. After using this tool, the authors reviewed and edited the
 1047 content as needed and take full responsibility for the content of the manuscript.

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